TEXTBOOK · FALL 2026

Math I

Weeks 1–4

All questions and all formulas from every course file and the assigned book sections — number systems, notation, complex numbers; functions; limits and limit laws, standard limits, indeterminate forms, asymptotics and the Landau symbols — each problem with a complete step-by-step solution. Compiled for independent study and Test 1 preparation

Tashkent Technological University · Fall 2026

Sources: course notes, lectures, ISA, practical sessions; OpenStax Calculus 1

QUESTIONBANK · FORMULACATALOG · FULLSOLUTIONSOCTOBER 2026

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1 Course Map and How to Use This Document

This document is a complete standalone guide for the first four weeks of the Math I course in the first semester (Fall 2026). It was manually compiled from each relevant source in the course-files/ folder of the course repository: weekly notes (Notes), lecture presentations, practical exercise sheets, independent study assignments (ISA), and designated sections of the OpenStax Calculus Volume 1 textbook. Problems are reproduced verbatim; solutions are complete, step-by-step, with verifications.

Week	Math I
1	Number sets $N \subseteq Z \subseteq Q \subseteq R \subseteq C$; positional systems; scientific and engineering notation; complex numbers
2	Functions: classes, domain/range, symmetry, composition, inverse functions; exponential and logarithmic functions (§1.2, §1.5)
3	Limits: intuitive concept, one-sided and infinite limits, asymptotes, precise $\varepsilon - \delta$ definition (§2.2, §2.5)
4	Limits II: limit laws, standard limits, indeterminate forms, squeeze theorem, asymptotic equivalence, Landau symbols O and o(§2.3)

Notation. Powers and indices are written using ^{} and _: $x^2, a_{11}, f^{(-1)}$. Sets are written conventionally: $N \subseteq Z \subseteq Q \subseteq R \subseteq C$. The derivative is denoted $f'(x)$ or dy/dx.

2 Week 1 — Number systems, notation, complex numbers (Math I)

2.1 Math I: Formula catalog for week 1

2.1.1 Number sets and positional systems

The nesting of the standard number sets is the first structural fact of the course, and every later classification exercise relies on it. Rational numbers are exactly the fractions of integers, while irrational numbers are the real numbers that are not rational.

$$N \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}, \quad \mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\} \text{ (irrationals)} \tag{1}$$

$$q \in \mathbb{Q} \Leftrightarrow q = \frac{a}{b},\ a \in \mathbb{Z},\ b \in \mathbb{Z},\ b \neq 0, \quad d(a,b) = |a-b| \text{ (distance in } \mathbb{R}) \tag{2}$$

A positional numeral system with base b writes a number through powers of the base; the same number admits many numerals, but its value is independent of the representation. Base conversion uses the expansion below or repeated division for the reverse direction.

$$(a_n a_{n-1} \ldots a_1 a_0)_b = \sum_{k=0}^{n} a_k b^k, \quad 0 \leq a_k \leq b-1 \tag{3}$$

$$13_{10} = 1101_2 = 15_8 = D_{16}, \quad A = 10,\ B = 11,\ \ldots,\ F = 15 \text{ (hexadecimal digits)} \tag{4}$$

Roman numerals are non-positional: $I = 1, V = 5, X = 10, L = 50, C = 100$, so $13 = XIII$; arithmetic in such systems is impractical, which is why computers use binary: electronics reliably distinguishes two states (low/high, off/on, 0/1).

2.1.2 Scientific and engineering notation

Scientific notation isolates the magnitude of a number into a coefficient with one nonzero digit before the decimal point and a power of ten; engineering notation constrains the exponent to multiples of three so that the exponent couples to an SI prefix.

$$\text{scientific: } z = a \times 10^{n}, \quad 1 \le |a| < 10,\ n \in \mathbb{Z} \tag{5}$$

$$\text{engineering: } z = a \times 10^{3k}, \quad 1 \le |a| < 1000,\ k \in \mathbb{Z} \tag{6}$$

$$(a \times 10^{m})(b \times 10^{n}) = ab \times 10^{m+n}, \quad \frac{a \times 10^{m}}{b \times 10^{n}} = \frac{a}{b} \times 10^{m-n} \tag{7}$$

$$(a \times 10^{n})^{m} = a^{m} \times 10^{nm}, \quad \text{add/subtract: rewrite to a common power of ten first} \tag{8}$$

The SI prefixes used all semester are:

Prefix	Symbol	Power
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
milli	m	10^{-3}
micro	µ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}

Case matters: $5\ \text{mW} = 5 \times 10^{-3} W$ but $5 MW = 5 \times 10^{6} W$. Reference values used throughout the course:

$$c = 2.99792458 \times 10^{8}\ \text{m/s}; \quad e = 1.602176634 \times 10^{-19}\ \text{C}; \quad N_A = 6.02214076 \times 10^{23}\ \text{mol}^{-1};$$

$$h = 6.62607015 \times 10^{-34}\ \text{J s}; \quad 101325\ \text{Pa} = 101.325\ \text{kPa}.$$

2.1.3 Complex numbers

Week 1 closes with the complex numbers, which arise because $x^2 + 1 = 0$ has no real solution. The imaginary unit, equality rule, the four arithmetic operations and the conjugate are the working tools.

$$i = \sqrt{-1},\ i^2 = -1,\ z = a + bi,\ \text{Re}(z) = a,\ \text{Im}(z) = b \tag{9}$$

$$a+bi=c+di \Leftrightarrow a=c \text{ and } b=d, \quad i^{n+4}=i^n \ (i^0=1,\ i^1=i,\ i^2=-1,\ i^3=-i) \tag{10}$$

$$z_1+z_2=(a+c)+(b+d)i, \quad z_1-z_2=(a-c)+(b-d)i \tag{11}$$

$$(a+bi)(c+di)=(ac-bd)+(ad+bc)i, \quad \frac{z_1}{z_2}=\frac{z_1\overline{z_2}}{|z_2|^2} \ (z_2\neq 0) \tag{12}$$

$$\bar{z}=a-bi,\ z+\bar{z}=2\,\mathrm{Re}(z),\ z-\bar{z}=2i\,\mathrm{Im}(z),\ z\bar{z}=a^2+b^2=|z|^2 \tag{13}$$

$$|z|=\sqrt{a^2+b^2}, \quad |z_1z_2|=|z_1|\,|z_2|, \quad \left|\frac{z_1}{z_2}\right|=\frac{|z_1|}{|z_2|} \tag{14}$$

$$|z_1+z_2|\leq |z_1|+|z_2| \text{ (triangle inequality)} \tag{15}$$

$$\arg z=\theta+2k\pi,\ k\in\mathbb{Z}; \quad \mathrm{Arg}(z)\in(-\pi,\pi] \text{ principal value, } \tan\theta=\frac{b}{a} \text{ (quadrant!)} \tag{16}$$

$$z=r(\cos\theta+i\sin\theta)=re^{i\theta}, \quad r=|z|,\ a=r\cos\theta,\ b=r\sin\theta \tag{17}$$

$$e^{i\theta}=\cos\theta+i\sin\theta \text{ (Euler)}, \quad e^{i\pi}+1=0 \tag{18}$$

$$z_1z_2=r_1r_2\,e^{i(\theta_1+\theta_2)}, \quad \frac{z_1}{z_2}=\frac{r_1}{r_2}\,e^{i(\theta_1-\theta_2)} \tag{19}$$

$$z^n=r^n\big(\cos(n\theta)+i\sin(n\theta)\big) \text{ (de Moivre)} \tag{20}$$

$$z_k=r^{1/n}\left(\cos\frac{\theta+2k\pi}{n}+i\sin\frac{\theta+2k\pi}{n}\right), \quad k=0,\dots,n-1 \tag{21}$$

$$z^n=1\Rightarrow z_k=e^{2\pi ik/n}, \quad k=0,\dots,n-1 \ (n\text{-th roots of unity}) \tag{22}$$

$$z=\frac{-b\pm\sqrt{b^2-4ac}}{2a} \text{ (quadratic formula, valid over } \mathbb{C}) \tag{23}$$

The roots of $z^n=w=\mathrm{re}^{i\theta}$ are given by (21); the $n-$th roots of unity form a regular $n-$gon on the unit circle. Non-real roots of real polynomials occur in conjugate pairs, and by the Fundamental Theorem of Algebra a degree$-n$ polynomial has exactly n roots in C, counted with multiplicity.

2.2 Math I: Week 1 Practical Session — questions and complete solutions

1. Classify each decimal as terminating, periodic, or non-periodic, and state whether the corresponding number is rational: (1) 0.375; (2) 0.272727 . . .; (3) 1.101001000100001 . . .; (4) 3.14159265 . . .; (5) 0.123123123 . . .

[Source: Week 1, practical session, exercise 1]

Solution. (1) 0.375 is finite, therefore rational: $0.375 = \frac{375}{1000} = \frac{3}{8}$. (2) Block 27 repeats infinitely, therefore rational; for a purely periodic block ab $= \frac{ab}{99}$, so $0.27 = 27_{99} = 3_{11}$. (3) The intervals between consecutive ones grow (1, 2, 3, . . . zeros), so the fraction will never become periodic or terminate: the number is irrational. $(4)A$ finite number of digits shown with ellipsis does not define the number; if $\pi = 3.14159265\ldots$ was intended, it is infinite and non-periodic, therefore irrational. (5) Block 123 repeats, therefore rational: $0.123 =_{333} =$. Principle: finite or eventually periodic decimal fractions are exactly the[41] rational numbers.

2. Write each set in interval notation and represent it on the number line: (1) {$x \in R$: $-3 \leq x < 4$}; (2) {$x \in R$: $x \neq 3$}.

[Source: Week 1, practical session, exercise 2]

Solution. (1) The endpoint -3 is included (square bracket), 4 is excluded (parenthesis): [−3, 4); on the number line, this is shown as a filled point at -3 and an open point at 4, with the segment between them highlighted. (2) All real numbers except 3: $(-\infty, 3) \cup (3, \infty)$; on the number line, the entire axis is filled with an open point at 3. Infinity is never included, so parentheses are always used.

3. Write each interval in inequality form: a. [−2, 5); b. $(-\infty, 4]$; c.$(1, \infty)$.

[Source: Week 1, practical session, exercise 3]

Solution. a. $-2 \leq x < 5$ (left endpoint included, right endpoint excluded). b. $x \leq 4$ (unbounded to the left, right endpoint included). c. $x > 1$ (strictly greater than, since the parenthesis excludes 1). √

4. Classify as rational or irrational: a. $\sqrt{3}$; b. $\sqrt{25}$; c. $\sqrt{50}$; i. $(1+\sqrt{5})(1-\sqrt{5})$.

[Source: Week 1, practical session, ex. 4]

Solution. a. 3 is not a perfect square, therefore $\sqrt{3}$ is irrational. b. $\sqrt{25} = 5 \in Z \subseteq Q$, rational. c. $\sqrt{50} = 5\sqrt{2}$; since $\sqrt{2}$ is irrational, $5\sqrt{2}$ is irrational. i. Expanding: $(1+\sqrt{5})(1-\sqrt{5}) = 1-5 = -4$, rational. The product of conjugates always eliminates the radical.

5. Given $z_1 = 3+$ 2i and $z_2 = 1-$ 4i; compute $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1/z_2, writing each answer in the form $a+$ bi.

[Source: Week 1, practical session, ex. 5]

Solution. Addition and subtraction act on the real and imaginary parts separately: $z_1 + z_2 = (3+1) + (2-4)i = 4-$ 2i; $z_1 - z_2 = (3-1) + (2+4)i = 2+$6i. In the product, we use $i^2 = -1$ and the rule $(a+$ bi$)(c+$ di) = (ac − bd) + (ad + bc)i: $z_1 z_2 = (3 \cdot 1 - 2 \cdot (-4)) + (3 \cdot (-4) + 2 \cdot 1)i = (3+8) + (-12+2)i = 11-$ 10i. In the quotient, we multiply numerator and denominator by the conjugate of the denominator:

$$(z_1)/(z_2) = ((3+2i)(1+4i))/((1-4i)(1+4i))$$
$$(3+12i+2i+8i^2)/(1+16) = ((3-8) + (12+$$

$$2)i)/17 = (-5 + 14i)/17 = -\frac{5}{17} + (\frac{14}{17})i.$$

6. For the numbers $3+$ 4i, $-2+$ 5i, 6i, −7, find the conjugate and modulus; verify $z\bar{z} = |z|^2$. $\sqrt{a^2 + b^2}$. $3+$ 4i: $\bar{z} = 3-$ 4i, $|z| = 5$,

[Source: Week 1, practical session, ex. 6]

Solution. For $z = a+$ bi we have $\bar{z} = a-$ bi and $|z| =$

$$\sqrt{a}^2 + b^2$$

. $-2+$ 5i: $\bar{z} = -2-$ 5i, $|z| = \sqrt{4 + 25} = \sqrt{29}$ and (3 − 4i)(3+ 4i) $= 9 + 16 = 25 = 5^2$. 29, and

$$6i : \bar{z} = -6i, |z| = 6, \text{and} (-6i)(6i) = -36i^2 = 36.$$

(−2 − 5i)(−2+ 5i) $= 4 + 25 = 29$. −7: $\bar{z} = -7$ (real numbers are conjugates of themselves), $|z| = 7, (-7)(-7) = 49$. All four checks agree.

1

7. Express $1/(1 + i)$ in the form $a+$ bi.

[Source: Week 1, practical session, ex. 7]

Solution. Multiply by the conjugate of the denominator:

$$1/(1 + i) = (1 - i)/((1 + i)(1 - i)) = (1 - i)/2 = \frac{1}{2} - i/2$$

8. Solve over C: $x^2 + 9 = 0$.

[Source: Week 1, practical session, ex. 8]

Solution. Rearranging, we get $x^2 = -9$, so $x = \pm\sqrt{-9} = \pm$3i. Both solutions are non-real complex numbers; there are no real solutions because a square in R is never negative.

9. For $z = 2+$ 3i on the complex plane, determine the real part, imaginary part, modulus, and quadrant. What changes for 2 − 3i?

[Source: Week 1, practical session, ex. 9]

Solution. The point (2, 3) lies in the first quadrant: $Re(z) = 2, Im(z) = 3, |z| =$

$$\sqrt{2^2 + 3^2} = \sqrt{1}3$$

. For $z = 2-$ 3i, the point (2, −3) is the reflection across the real axis: the real part and modulus do not change, the imaginary part changes sign, and the point moves to the fourth quadrant.

10. Write $1 + i$ in trigonometric form $z = r(\cos\theta + i\sin\theta)$ and exponential form $z =$ re$^{i\theta}$, with θ in radians.

[Source: Week 1, practical session, ex. 10]

Solution. Here $r = |1 + i| = \sqrt{1} + 1 = \sqrt{2}$, both coordinates are positive, so $\theta = arctan(\frac{1}{1}) = \pi/4$. Thus, $1 + i = \sqrt{2}(\cos\pi/4 + i\sin\pi/4) = \sqrt{2}e^{i\pi/4}$.

11. Using De Moivre's formula $[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta)$, compute $(\sqrt{3} + i)^4$; verify algebraically. $|\sqrt{3} + i| = \sqrt{3 + 1} = 2$, $\theta = arctan(1/\sqrt{3}) = \pi/6$.

[Source: Week 1, practical session, ex. 11]

Solution. First, convert: $r = |\sqrt{3}+i| = \sqrt{3+1} = 2$, $\theta = arctan(1/\sqrt{3}) = \pi/6$. By De Moivre's,

$$(\sqrt{3}+i)^4 = 2^4(\cos(4\arctan(1/\sqrt{3})) + i\sin(4\arctan(1/\sqrt{3})))$$
$$16(\cos(4\pi/6) + i\sin(4\pi/6)) =$$

$$16(\cos(2\pi/3) + i\sin(2\pi/3)) = 16(-\frac{1}{2} + \sqrt{\frac{3}{2}}i) = -8 + 8\sqrt{3}i.$$

Algebraic verification:

$$(\sqrt{3}+i)^2 = 3 + 2\sqrt{3}i + i^2 = 2 + 2\sqrt{3}i;$$

squaring again,

$$(2+2\sqrt{3}i)^2 = 4 + 8\sqrt{3}i + 12i^2 = -8 + 8\sqrt{3}i.$$

Both methods agree.

12. Find all solutions to $z^3 = -8$ and plot them on the complex plane.

[Source: Week 1, practical session, exercise 12]

Solution. We write $-8 = 8e^{i(\pi+2k\pi)}$. The cube roots have modulus $8^{1/3} = 2$ and arguments $(\pi+2k\pi)/3, k = 0, 1, 2$:

$$z_0 = 2e^{i\pi/3} = 1 + \sqrt{3}i, z_1 = 2e^{i\pi} = -2,$$
$$z_2 = 2e^{i5\pi/3} = 1 - \sqrt{3}i.$$

Geometrically, these are three equally spaced points on a circle of radius 2 centered at the origin (angles $60^\circ, 180^\circ, 300^\circ$); note that $z_1 = -2$ is the only real solution, which is consistent with $(-2)^3 = -8$.

13. For each number, specify the smallest set from N, Z, Q, R, C to which it belongs: 0, -7, $\frac{5}{8}$, 3.25, $\sqrt{16}$, $\sqrt{7}$, $-\sqrt{9}$, π, π^2, i, $4-$ **3i.**

[Source: Week 1, ISA, assignment 2]

Solution. $0 \in N$(by course convention) ⇒ smallest set N. $-7 \in Z$. 5/8 and $3.25 = \frac{13}{4}$ lie in Q (finite decimal fractions). $\sqrt{16} = 4 \in N$. $\sqrt{7}$ is irrational (7 is not a perfect square) $\Rightarrow R$. $-\sqrt{9} = -3 \in Z$. π and π^2 are irrational $\Rightarrow R$ (squaring does not make an irrational number rational). i and 4 − 3i are non-real ⇒ only C.

14. True or false? Correct each false statement: $(1) N \subseteq Z$**;** $(2) Z \subseteq Q$**;** $(3) R \subseteq Q$**;** $(4) Q \subseteq R$**;** $(5) R \subseteq C$**; (6) every integer is natural; (7) every rational number is real; (8) every complex number is real; (9) every real number can be written as a complex number;** $(10) a$ **number cannot be both rational and real.**

[Source: Week 1, ISA, assignment 3]

Solution. (1) True. (2) True. (3) False — for example, $\sqrt{2} \in R$, but $\sqrt{2} \notin Q$; the correct inclusion is $Q \subseteq R$. (4) True. (5) True. (6) False — negative integers,

for example -3, are not natural. (7) True. (8) False — for example, i is complex but not real. (9) True — $a = a+$ 0i. (10) False — every rational number is real (statement (4)), so rational numbers are both.

15. Draw a diagram of nested sets $N \subset Z \subset Q \subset R \subset C$ and place $5,\ -2,\ \frac{2}{3},\ \sqrt{2},\ \pi,\ -4\text{i},\ 3+2\text{i}$ in the correct regions.

[Source: Week 1, ISA, assignment 4]

Solution. Draw five nested ovals labeled N (innermost) to C (outermost). Placement: $5 \in N$; $-2 \in Z \ \ N$; $\frac{2}{3} \in Q \ \ Z$; $\sqrt{2}$ and $\pi \in R \ \ Q$; -4i and $3+2\text{i} \in C \ \ R$. The diagram visually shows that each new set only adds numbers, removing nothing.

16. Arrange in increasing order: $-\sqrt{5},\ -2,\ -\frac{3}{2},\ 0,\ \sqrt{2},\ \frac{3}{2},\ \pi/2$; justify non-obvious comparisons.

$$(\frac{2}{3})^2 = \frac{4}{9} < (\pi/2)^2$$

[Source: Week 1, ISA, assignment 5]

Solution. Squares give the order for positive numbers: $\sqrt{\frac{3}{4}} = (\sqrt{3})/2 \approx 0.866$; since $\pi/2 \approx 1.5708$ and $\frac{3}{2} = 1.5$, we have $\frac{3}{2} < \pi/2$. For negative numbers the order is reversed: $-\sqrt{5} \approx -2.236 < -2 < -\frac{3}{2}$. In total: $-\sqrt{5} < -2 < -\frac{3}{2} < (\sqrt{3})/2 < \frac{3}{2} < \pi/2$.

$$-\sqrt{5} < -2 < -\frac{3}{2} < 0 < 2 < \frac{3}{2} < \pi/2$$

17. Write each set in interval notation: $(1) -3 \le x < 4$; $(2) x > 2$; $(3) x \le -1$; $(4) -2 < x \le 5$; $(5) x \ne 3$; then graph each on the number line.

[Source: Week 1, ISA, assignment 6]

Solution. (1) $[-3,\ 4)$. $(2)(2, \infty)$. $(3)(-\infty,\ -1]$.$(4)(-2,\ 5]$.$(5)(-\infty,\ 3) \cup (3,\ \infty)$. On the number line: filled points where the endpoint is included (≤, ≥), open points where it is not (<, >), parentheses with infinities, and two rays for the union in (5).

18. Write each interval as an inequality: a. $[-2,\ 5)$; b. $(-\infty,\ 4]$; $c.(1,\ \infty)$; $d.(-3,\ 0] \cup [2,\ 6)$; $e.R\ [-1,\ 1]$.

[Source: Week 1, ISA, assignment 7]

Solution. $a.\ -2 \le x < 5$. $b.\ x \le 4$. $c.\ x > 1$. $d.\ -3 < x \le 0$ or $2 \le x < 6$. $e.\ x < -1$ or $x > 1$; equivalently $|x| > 1$.

19. With $d(a,b) = |a-b|$: compute $d(3,\ 8),\ d(-2,\ 5),\ d(-4,\ -9),\ d(2,\ 8)$; check $d(a,\ c) \le d(a,\ b) + d(b,\ c)$ for $a = -3, b = 2, c = 7$.

$$d(2,8) = |2-8|/2 = 2$$

[Source: Week 1, ISA, assignment 8]

Solution. $d(3,8) = |3-8| = 5$; $d(-2,5) = 7$; $d(-4,-9) = 5$; $d(2,8) = |2-8|/2 = 2$. Checking the triangle inequality: $d(-3,\ 7) = 10$, $d(-3,\ 2) + d(2,\ 7) = 5 + 5 = 10$, and indeed $10 \le 10$ (equality, since the points lie in order on a straight line). The formula $|a-b|$ is the one-dimensional case of the *norm* from (28).

20. Determine whether each number is rational or irrational; justify: a. $\sqrt{49}$; b. $\sqrt{18}$; c. $\sqrt{3}+\sqrt{2}$; d. $\sqrt{2}\cdot\sqrt{5}$; e. $\sqrt{2}+\sqrt{8}$; f. $(\sqrt{3})^2$; g. $(1+\sqrt{2})(1-\sqrt{2})$; h. $\sqrt{12}$.

[Source: Week 1, ISA, assignment 9]

Solution. a. $\sqrt{49} = 7$ — rational. b. $\sqrt{18} = 3\sqrt{2}$ — irrational. c. $\sqrt{3}+\sqrt{2}$ — irrational. d. $\sqrt{2}\cdot\sqrt{5} = \sqrt{10}$ — irrational. e. $\sqrt{2}+\sqrt{8} = \sqrt{2}+2\sqrt{2} = 3\sqrt{2}$ — irrational. f. $(\sqrt{3})^2 = 3$ — rational. g. $(1+\sqrt{2})(1-\sqrt{2}) = 1-2 = -1$ — rational (product of conjugates). h. $\sqrt{12} = 2\sqrt{3}$ — irrational. Practical rule: simplification reveals nature; products of conjugates of the form $(a+\sqrt{b})(a-\sqrt{b})$ are always rational.

21. Given $z_1 = 3+$ 2i, $z_2 = 1-$ 4i; compute $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1/z_2.

[Source: Week 1, ISA, assignment 10]

Solution. Same as exercise 5 of the practical session above: $z_1 + z_2 = 4-$ 2i, $z_1 - z_2 = 2+$ 6i, $z_1 z_2 = 11-$ 10i, and $z^1 = -175 + 1417i$.

$$z_2 = 1 - 4i$$

22. Compute i^2, i^3, i^4, i^5, i^{18}, i^{27}, i^{100}; describe the repeating pattern and the method based on dividing the exponent by 4.

[Source: Week 1, ISA, assignment 11]

Solution. $i^2 = -1$; $i^3 = i^2 i = -i$; $i^4 = (i^2)^2 = 1$; $i^5 = i$. Powers are cyclic with period 4: $1, i, -1, -i, \ldots$ For a large exponent, divide by 4 with remainder r: $i^n = i^r$. Then $18 = 4\cdot 4 + 2 \Rightarrow i^{18} = i^2 = -1$; $27 = 4\cdot 6 + 3 \Rightarrow i^{27} = i^3 = -i$; $100 = 4\cdot 25 \Rightarrow i^{100} = i^0 = 1$.

23. For $3+$ 4i, $-2+$ 5i, 6i, -7: find the conjugate and modulus, verify $z\bar{z} = |z|^2$.

[Source: Week 1, ISA, assignment 12]

Solution. Same calculations as in exercise 6 of the practical session: √ conjugates $3 - 4$i, $-2 - 5$i, -6i, -7; moduli 5, 29, 6, 7; in each case $z\bar{z} = a^2 + b^2 = |z|^2$ holds exactly.

24. Express in the form $a+$ bi: a. $1/(1+i)$; b. $3+2/(2-i)$; c. $1+3/(4-i)$; d. $5/(2-i)$.

[Source: Week 1, ISA, assignment 13]

Solution. In each case, multiply numerator and denominator by the conjugate of the denominator. a.

$$(1+i)/(1-i) = (1+i)^2/((1-i)(1+i)) = (1+2i+i^2)/(1-i^2)$$
$$(1+2i-1)/(1-(-1)) = (2i)/2 = i$$

$= i$. b.

$$(3+2i)/(2-i) = (3+2i)(2+i)/((2-i)(2+i)) = (6+3i+4i+2i^2)/(4-i^2)$$
$$(6+7i-2)/(4-(-1)) = (4+7i)/5$$

$= \frac{4}{5} + \frac{7}{5}i$. c.

$$(1+3i)/(4-i) = (1+3i)(4+i)/((4-i)(4+i)) = (4+i+12i+3i^2)/(16-i^2)$$
$$(4+13i-3)/(16-(-1)) = (1+13i)/17$$

$= \frac{1}{17} + \frac{13}{17}i$. d.

$$5i/(2-i) = 5i(2+i)/((2-i)(2+i)) = (10i+5i^2)/(4-i^2)$$
$$(10i-5)/(4-(-1)) = (-5+10i)/5$$

$= -1+$ 2i.

25. Solve over C, indicating which solutions are real: a**.** $x^2+9=0$**;** b**.** x^2- **4x** $+13=0$**;** c**.** **2x**$^2+$ **4x** $+10=0$**;** d**.** $x^4-16=0$**.**

[Source: Week 1, ISA, assignment 14]

Solution. a. $x = \pm$3i (non-real). b. Discriminant $\Delta = (-4)^2 - 4\cdot 13 = 16-52 = -36$, $\sqrt{\Delta}$ = 6i; $x = (4\pm$ 6i$)/2 = 2\pm$ 3i (non-real). c. Divide by 2:

$$x = (-4 \pm \sqrt{4^2 - 4\cdot 2\cdot 10})/(2\cdot 2) = (-4\pm\sqrt{-72})/4 = (-4\pm 6i\sqrt{2})/4 = -1 \pm (3i\sqrt{2})/2$$

x^2+ 2x $+5=0$: $\Delta = 4-20 = -16$, $\sqrt{\Delta}$ = 4i; $x = (-2\pm$ 4i$)/2 = -1\pm$ 2i (non-real).

d. $x^4 = 16$ gives real solutions $x = \pm 2$; over C the complete set is the four fourth roots of 16: 2, -2, 2i, -2i (all solutions $(x^2-4)(x^2+4)=0$).

26. For $2+$**3i,** $-1+$**4i, −3 − 2i,** $4-i$**: find the real part, imaginary part, modulus, quadrant. √**

[Source: Week 1, ISA, assignment 15]

Solution. $2+$ 3i: (2, 3), modulus 13, quadrant I. $-1+$ 4i: (−1, 4), modulus 17, $\sqrt{}$ quadrant II. $-3-$ 2i: (−3, −2), modulus 13, quadrant III. $4-i$: (4, −1), modulus 17, quadrant IV. Note that moduli pair up, since conjugation preserves a^2+b^2.

27. Write in trigonometric and exponential forms (θ **in radians):** $(1)\sqrt{1}+i$**;** $(2)-1+i$**;** $(3)-3-i$**; (4) 2i; (5) −4. √**

[Source: Week 1, ISA, assignment 16]

Solution. $(1) r = 2, \theta = \pi/4$: 2e$^{i\pi/4}$. $(2) r = 2$, quadrant II: $\theta = 3\pi/4$: 2e$^{i3\pi/4}$. $(3) r = \sqrt{3+1} = 2$, quadrant III: reference angle $arctan(1/\sqrt{3}) = \pi/6$, so $\theta = \pi+\pi/6 = 7\pi/6$: 2e$^{i7\pi/6}$. $(4) r = 2, \theta = \pi/2$: 2e$^{i\pi/2}$. $(5) r = 4, \theta = \pi$: 4e$^{i\pi}$. Each angle is taken from $\tan\theta = b/a$ with adjustment for quadrant, as warned in (16). √

28. Using De **Moivre's formula, calculate** $(1)(1+i)^6$**;** $(2)(3+i)^4$**;** $(3)(-1+3i)^5$**; check one result algebraically. √**

$$(1+i)^6 = (\sqrt{2})^6 e^{i6\pi/4} = 8e^{i3\pi/2} = 8(0-i) = -8i.$$

[Source: Week 1, ISA, assignment 17]

Solution. $(1) 1+i = \sqrt{2}e^{i\pi/4}$. Then $(1+i)^6 = (\sqrt{2})^6 e^{i6\pi/4} = 8e^{i3\pi/2} = 8(0-i) = -8$i. (2) From exercise 11 of the practical session: $(3+i)^4 = -8+8\sqrt{3}i$. $(3) z = -1+$ 3i: modulus $|z| = \sqrt{1+3} = 2$; quadrant II, reference angle $\pi/3$, so $\theta = 2\pi/3$: $z =$ 2e$^{i2\pi/3}$. Then $z^5 = 2^5 e^{i10\pi/3} =$ 32e$^{i(10\pi/3-2\pi)} =$ 32e$^{i4\pi/3} = 32(-\frac{1}{2} - i\sqrt{3}/2) = -16-16\sqrt{3}i$. Verification (1) algebraically: $(1+i)^2 =$ 2i, $(1+i)^4 = ($2i$)^2 = -4, (1+i)^6 = -4\cdot$ 2i = -8i. Matches.

29. Find all solutions and plot them on the complex plane: a**.** $z^3=1$**;** b**.** $z^4=1$**;** c**.** $z^3=-8$**. Explain why the** $n-$**th roots are equidistant on** a **circle. √**

[Source: Week 1, ISA, assignment 18]

Solution. a. By (22): $z_k = e^{2\pi ik/3}, k = 0,1,2$: $1, -\frac{1}{2}+\frac{3}{2}i, -\frac{1}{2}-\frac{3}{2}i$. b. $z_k = e^{2\pi ik/4}$: $1, i, -1, -i$. c. From exercise 12 of the practical session: $\sqrt{1}+$ 3i, -2, 1 − 3i. In

all cases, the roots have the same modulus $r^{1/n}$ (thus lie on the same circle), and their arguments differ by exactly $\frac{2\pi}{n}$(a full turn 2π is divided into n equal arcs) — that's why they form the vertices of a regular $n-$gon.

30. The electric charge $e = 0.000000000000000000001602176634C$. Write it in scientific and engineering notation.

[Source: Week 1, ISA, assignment 19]

Solution. We count 18 digits so that the first non-zero digit is before the decimal point: $e = 1.602176634 \times 10^{-19}C$ (scientific). In engineering notation, the exponent must be a multiple of 3, so we shift by one digit: $e = 160.2176634 \times 10^{-21}C = 160.2176634$ zC (zeptocoulombs); in practice, they write $160.2176634 \times 10^{-21}C$, since below 10^{-24} there is only zepto (10^{-21}) among the standard SI prefixes.

31. Fill in the table of SI prefixes: powers 10^{12}, 10^{6}, 10^{3}, 10^{-6}, 10^{-9}, 10^{-12} with prefixes/notations.

[Source: Week 1, ISA, assignment 20]

Solution. 10^{12} tera (T); 10^{6} mega (M); 10^{3} kilo (k); 10^{-6} micro (µ); 10^{-9} nano (n); 10^{-12} pico (p). Together with giga (G), milli (m), femto (f), atto (a), this table forms the complete SI ladder.

32. Week 1 summary, part 2, in-class problems: (1) write 0.00000824 in scientific and engineering notation; (2) write $7.35 \times 10^8 Hz$ in engineering notation; (3) write $0.0047F$ with an SI prefix.

[Source: Week 1 summary, part 2]

Solution. $(1) 0.00000824 = 8.24 \times 10^{-6}$ (scientific); since -6 is a multiple of 3, this is already in engineering notation, for example 8.24 µA if the unit is ampere. $(2) 10^8$ is not a multiple of 3; we rewrite $7.35 \times 10^8 = 735 \times 10^6 = 735$ MHz. $(3) 0.0047 = 4.7 \times 10^{-3} = 4.7$ mF.

33. Week 1 summary, part 3, in-class problems: (1) draw a diagram of nested sets $N \subset Z \subset Q \subset R \subset C$ and place 2026, -9, $\frac{2}{3}$, 10, π, -5i, $-2+2$i; (2) rational or irrational: a. $\sqrt{49}, b$. $\sqrt{18}, c$. $\sqrt{3}+\sqrt{2}, d$. $\sqrt{2}\cdot\sqrt{5}, e$. $\sqrt{2}+\sqrt{8}, f$. $(\sqrt{5})^2$.

[Source: Week 1 summary, part 3]

Solution. $(1) 2026 \in N$; $-9 \in Z \setminus N$; $\frac{2}{3} \in Q \setminus Z$; 10 and $\pi \in R \setminus Q$; -5i and $-2+2$i $\in C \setminus R$. $(2) a$. $\sqrt{49} = 7$ — rational. b. $\sqrt{18} = 3\sqrt{2}$ — irrational. c. $\sqrt{3}+\sqrt{2}$ — irrational. d. $\sqrt{2}\cdot\sqrt{5} = \sqrt{10}$ — irrational. e. $\sqrt{2}+\sqrt{8} = 3\sqrt{2}$ — irrational. f. $(\sqrt{5})^2 = 5$ — rational — same logic as in assignment 9, which this exercise repeats with different numbers.

34. Week 1 summary, part 3, in-class problems: given $z_1 = 2+3$i, $z_2 = 1-5$i; calculate $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1/z_2.

[Source: Week 1 summary, part 3]

Solution. $z_1 + z_2 = 3-2$i; $z_1 - z_2 = 1+8$i; product: $(2\cdot 1 - 3\cdot(-5)) + (2\cdot(-5) + 3\cdot 1)i = (2+15) + (-10+3)i = 17-7$i; quotient:

$$(z_1)/(z_2) = ((2+3i)(1+5i))/((1-5i)(1+5i))$$
$$(2+10i+3i+15i^2)/(1-(5i)^2) = (2+13i-15)/(1+$$

$$25) = (-13+13i)/26 = -\frac{1}{2} + \frac{1}{2}i.$$

35. Explain why $1+i\sqrt{3}=2(\cos\pi/3+i\sin\pi/3)=\mathbf{2e}^{i\pi/3}$**.**

$$|1+i\sqrt{3}|=\sqrt{1^2+(\sqrt{3})^2}=\sqrt{1+3}=2.$$

[Source: Week 1, notes, part 3]

Solution. The modulus is $|1+i\sqrt{3}|$. The point $(1,\sqrt{3})$ lies in the first quadrant, and $\tan\theta=\sqrt{3}$, from which $\theta=\pi/3$. Substituting $r=2,\theta=\pi/3$ into the polar form (17), and then into Euler's formula (18), we obtain both equalities.

3 Week 2 — Functions (Math I)

3.1 Math I: Formula catalog for week 2

3.1.1 Functions: definition, classes, symmetry, composition, inverses

A function assigns to each x in its domain exactly one output $f(x)$; the vertical-line test detects relations that fail this. The natural domain is the largest set of real inputs for which the formula makes sense; the three standard obstructions are division by zero, even roots of negatives, and logarithms of nonpositive numbers.

$$f:D\to\mathbb{R},\ x\mapsto f(x);\quad D=\text{domain},\ \text{range}\,f=\{f(x):x\in D\} \tag{31}$$

$$\frac{1}{g(x)}:g(x)\neq 0,\quad \sqrt{g(x)}:g(x)\geq 0,\quad \log_b g(x):g(x)>0 \tag{32}$$

Evenness and oddness reflect symmetry about the $y-$axis and about the origin respectively; composition chains functions; only one-to-one functions (horizontal-line test) possess inverses, recovered by solving $y=f(x)$ for x.

$$\text{even: } f(-x)=f(x),\quad \text{odd: } f(-x)=-f(x) \tag{33}$$

$$(f\circ g)(x)=f\big(g(x)\big),\quad (f\circ g)\neq(g\circ f)\text{ in general} \tag{34}$$

$$f\text{ one-to-one}\Leftrightarrow f(x_1)=f(x_2)\Rightarrow x_1=x_2 \tag{35}$$

$$f^{-1}\big(f(x)\big)=x\text{ and }f\big(f^{-1}(y)\big)=y \tag{36}$$

$$f^{-1}:\ \text{graph}=\text{mirror of the graph of }f\text{ in }y=x \tag{37}$$

The function classes of §1.2 (OpenStax) are linear, polynomial, rational, root (algebraic), exponential/logarithmic and trigonometric (transcendental); every polynomial is smooth everywhere, and every rational function is defined wherever its denominator is nonzero.

3.1.2 Exponential and logarithmic functions (§1.5, hyperbolic functions excluded)

The exponential and logarithm are inverse to each other; the laws of exponents and logs convert multiplication into addition and are used in every growth/decay model of the course.

$$b^x > 0, \quad b^{x+y} = b^x b^y, \quad (b^x)^y = b^{xy}, \quad b^{-x} = \frac{1}{b^x} \tag{38}$$

$$\log_b x = y \Leftrightarrow b^y = x, \quad x > 0,\ b > 0,\ b \neq 1 \tag{39}$$

$$\log_b(xy) = \log_b x + \log_b y, \quad \log_b \frac{x}{y} = \log_b x - \log_b y, \quad \log_b x^n = n \log_b x \tag{40}$$

$$\log_b x = \frac{\ln x}{\ln b} \text{ (change of base)}, \quad b^x = e^{x \ln b} \tag{41}$$

$$e^{\ln x} = x, \quad \ln 1 = 0, \quad \ln e = 1 \tag{42}$$

Growth and decay models compare $A(t) = A_0 a^t$ with $A_0 e^{kt}$ through the growth-rate constant; the doubling time is the benchmark quantity.

$$A(t) = A_0 a^t = A_0 e^{kt}, \quad k = \ln a, \quad a > 1: \text{ growth}; \ 0 < a < 1: \text{ decay} \tag{43}$$

$$N(t) = N_0 2^{t/T_d} = N_0 e^{kt}, \quad k = \frac{\ln 2}{T_d}, \quad T_d = \frac{\ln 2}{k} \tag{44}$$

Degrees and radians convert through π rad $= 180°$:

$$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$$

3.2 Math I: Week 2 workshop (functions) — questions and full solutions

1. Determine whether each relation defines y as a function of x: a. $y = 3x -$

$$x^2 + y^2 = 9.$$

2; b. Where appropriate, solve for y and apply the vertical line test.

[Source: Week 2, workshop, exercise 1]

Solution. a. $y = 3x - 2$ gives exactly one y for each x: a function (a line). $\sqrt{b}$. Solving gives $y = \pm 9 - x^2$: most x(for example $x = 0$) give two y values $= \pm 3$, so a vertical line intersects the circle twice; not a function.

2. Let $f(x) = 2x^2 - 3x + 1$. Calculate and simplify $f(0),\ f(-2),\ f(a),\ f(a+h)$ and $f(a+h) - f(a), h \neq 0$. h

[Source: Week 2, workshop, exercise 2]

Solution. $f(0) = 1$; $f(-2) = 2(4) + 6 + 1 = 15$; $f(a) = 2a^2 - 3a + 1$; $f(a+h) = 2(a+h)^2 - 3(a+h) + 1 = 2a^2 + 4ah + 2h^2 - 3a - 3h + 1$. Difference quotient:

$$(f(a+h) - f(a))/h = (4ah + 2h^2 - 3h)/h = 4a + 2h - 3.$$

This expression anticipates the derivative of weeks 3–4 (as $h \to 0$ it approaches 4a − 3).

$$:_{x2+9}; b.g(x) = \sqrt{5} - 2x.$$

3. Find the natural domain: $a.\ f(x) =^{x-1}$ **Write in interval notation. The denominator** $x^2 + 9 \geq 9 > 0$ **never equals zero,**

[Source: Week 2, practical session, ex. 3]

Solution. $a.$ therefore the domain is all $R = (-\infty, \infty)$. $b.$ We need 5 − 2x $\geq 0 \Leftrightarrow \Rightarrow$ $x \leq \frac{5}{2}$: domain $(-\infty, \frac{5}{2}]$.

4. Let $f(x) =$ **2x** -3, $g(x) = x^2 + 1$**. Find** $(f \circ g)(x)$ **and** $(g \circ f)(x)$**; are they equal?**

[Source: Week 2, practical session, ex. 4]

Solution. $(f \circ g)(x) = f(g(x)) = 2(x^2 + 1) - 3 =$ 2x$^2 - 1$; $(g \circ f)(x) = g($2x $-3) = ($2x $-3)^2 +1 =$ 4x$^2 -$ 12x $+10$. They are different (for example, at $x = 0$: -1 vs 10), which confirms: composition is generally non-commutative — squaring then shifting is not the same as shifting then squaring.

5. Determine whether each function is even, odd, or of general form: $|x| + 1$ **and** $\frac{x}{x2+1}$**; check algebraically and graphically.**

[Source: Week 2, practical session, ex. 5]

Solution. $f(x) = |x| + 1$: $f(-x) = |-x| + 1 = |x| + 1 = f(x)$ — even; the graph is symmetric with respect to the $y-$axis. $g(x) = x/(x^2 + 1)$: $g(-x) = -x/(x^2 + 1) = -g(x)$ — odd; the graph is symmetric with respect to 180° rotation about the origin.

6. Classify each function as linear, polynomial, rational, algebraic, or transcendental (a function can $\sqrt{x+1}x^2 + 1$**. belong to more than one category):** $_{x2+4}$ **and**

[Source: Week 2, practical session, ex. 6]

Solution.$^{x+1}$ — a ratio of polynomials, hence rational (and

$$\sqrt{x^2 + 4}$$

$x^2 + 1$ — a root of a polynomial, hence algebraic algebraic). (not rational, as it is not a ratio of polynomials).

7. Temperature in an industrial chamber increases linearly from $20^\circ C$ **at** $t = 0$ **to** $80^\circ C$ **at** $t = 15\,\text{min}$**. (1) Find** $T(t)$**; (2) interpret the slope and intercept; (3) find** $T(8)$**; (4) when will the chamber reach** $60^\circ C$**? (5) specify** a **physically reasonable domain.**

[Source: Week 2, practical session, ex. 7]

Solution. (1) Slope $m = (80 - 20)/(15 - 0) = 4$ °$C/$ min, so $T(t) = 20+$ 4t. (2) The slope is the heating rate: $+4$ degrees per minute; the intercept 20 is the initial temperature. $(3)T(8) = 20 + 32 = 52$ °C. $(4)20+$ 4t $= 60$ at $t = 10$ minutes.

8. For $p(x) = x^3 -$ **4x determine: domain; intercepts with axes; symmetry; end behavior; points of discontinuity.**

[Source: Week 2, practical session, ex. 8]

Solution. Domain R (polynomial). Intercepts: $p(0) = 0$; $x^3 -$ 4x $= x(x - 2)(x + 2) = 0$ gives $x = -2, 0, 2$ — all on the $x-$axis through the origin. Symmetry: $p(-x) =$ $-x^3 +$4x $= -p(x)$ — odd. End behavior: the leading term x^3 with positive coefficient,

so $p(x) \to +\infty$ as $x \to +\infty$ and $p(x) \to -\infty$ as $x \to -\infty$. The polynomial is defined everywhere: no points of discontinuity.

9. An electricity supplier charges $C(E) = 0.10E$ for $0 \le E \le 200$ and $C(E) = 20 + 0.15(E - 200)$ for $E > 200$, in kWh. (1) Find $C(150)$, $C(200)$, $C(350)$; (2) graph it; (3) is the model continuous at $E = 200$? (4) meaning of each slope; (5) one limitation.

[Source: Week 2, practical session, ex. 9]

Solution. $(1) C(150) = 15$; $C(200) = 20$; $C(350) = 20 + 0.15 \cdot 150 = 42.5$(all in \$). (2) Two linear pieces: slope 0.10 up to 200, then 0.15 — a piecewise linear graph with a corner at (200, 20). (3) Yes: both formulas give 20 at $E = 200$, the pieces connect without a jump. (4) The slopes are marginal prices: 0.10\$/kWh for the first 200 kWh, 0.15\$/kWh beyond. (5) Limitation: the price jump exactly at 200 kWh incentivizes the consumer to stop just below the threshold; real tariffs change more smoothly or in blocks.

10. Find the inverse of $f(x) =$ 4x − 7 and specify the domains; verify $f(f(x)) = x$. $^{-1}$

[Source: Week 2, practical session, ex. 10]

Solution. $y = 4x - 7 \Rightarrow x = (y + 7)/4$, so $f^{-1}(x) = (x + 7)/4$ with domain R and range R. Check: $f(f^{-1}(x)) = 4 \cdot (x + 7)/4 - 7 = x + 7 - 7 = x$. Both are lines (the graphs are symmetric about $y = x$).

11. For each model, determine growth/decay and specify the initial value and growth/decay rate: $A(t) = 500(1.08)^t$, $B(t) = 1200(0.85)^t$, $C(t) =$ 40e$^{0.03t}$, $D(t) =$ 200e$^{-0.4t}$.

[Source: Week 2, practical session, ex. 11]

Solution. A: growth, initial 500, rate $1.08 (+8\%$ per period). B: decay, initial 1200, rate 0.85 (-15%). C: growth, initial 40, continuous rate $k = 0.03$. D: decay, initial 200, continuous rate $k = -0.4$. The sign of k in e^{kt} or whether $a > 1$ in a^t determines the direction.

12. Solve $2^{x+1} = 16$.

[Source: Week 2, practical session, ex. 12]

Solution. $16 = 2^4$, so $x + 1 = 4$ and $x = 3$. (Equivalently, take $\log_2$ of both sides.)

13. Battery discharge: $E(t) = E_0 e^{-0.025t}$, t in hours. (1) Fraction of remaining energy after $10h$? (2) When does 50% remain? (3) When does 10% remain? (4) Why is exactly zero unreachable?

[Source: Week 2, practical session, ex. 13]

Solution. $(1) E(10)/E_0 = e^{-0.25} \approx 0.7788$: about 77.9%. $(2) e^{-0.025t} = \frac{1}{2} \Rightarrow t = \ln \frac{2}{0}.025 \approx 27.7h$.

$(3) e^{-0.025t} = 0.1 \Rightarrow t = \ln \frac{10}{0}.025 \approx 92h$.

14. Population growth: $P(t) =$ 800e$^{0.35t}$, t in hours. (1) Initial population? (2) After $5h$? (3) Doubling time? (4) Can the model remain realistic indefinitely?

[Source: Week 2, practical session, ex. 14]

Solution. $(1) P(0) = 800$. $(2) P(5) =$ 800e$^{1.75} \approx 800 \cdot 5.755 \approx 4603$. $(3) T_d = \ln \frac{2}{0}.35 \approx 1.98h$. (4) No: exponential growth ignores nutrient depletion and space; realistically, the model works only near the start of the culture.

3.3 Math I: Week 2 homework problems and complete solutions

1. **Determine whether each relation defines y as a function of x: a. $y = 3x - 2$; b. $x^2 + y^2 = 9$; c. $y^2 = x + 1$; d. $x = y^3 - 2$.**
   [Source: Week 2, ISA, task 2]

   Solution. a. Function. b. Not a function (circle: two y for each x). c. $y = \pm\sqrt{x} + 1$: two values for $x > -1$, not a function. d. Solve for $\sqrt[3]{x} + 2$; cube root is unique for each real y: $y = x$, so it's a function (passes vertical line test, though not horizontal).

2. **For $f(x) = 2x^2 - 3x + 1$, calculate $f(0)$, $f(-2)$, $f(a)$, $f(a+h)$, $\frac{f(a+h)-f(a)}{h}$.**
   [Source: Week 2, ISA, task 3]

   Solution. Same as exercise 2 of the practical session: 1; 15; $2a^2 - 3a + 1$; $2a^2 + 4ah + 2h^2 - 3a - 3h + 1$; difference quotient = $4a + 2h - 3$.

   $$(f(a+h) - f(a))/h = 4a + 2h - 3.$$

3. **Find the natural domain: a. $f(x) = \frac{x-1}{q}$**

   $$\sqrt{(x+2)/(x-1)}$$

   c. $h(x) = \sqrt{(x-1)/(x-4)}$; d. $p(x) = \sqrt{(x+2)/(x-1)}$.
   [Source: Week 2, ISA, task 4]

   Solution. a. $(-\infty, \infty)$. b. $(-\infty, \frac{5}{2}]$.c. Need $x - 1 \geq 0$ and $x \neq 4$: $[1, 4) \cup (4, \infty)$. d. Need $\frac{x+2}{x-1} \geq 0$ and $x \neq 1$: critical points -2 and 1; quotient non-negative on $(-\infty, -2] \cup (1, \infty)$.

4. **Find the range: $f(x) = x^2 + 2$, $g(x) = -|x - 1| + 4$, $h(x) = \sqrt{x} + 3 - 2$.**
   [Source: Week 2, ISA, task 5]

   Solution. $x^2 \geq 0 \Rightarrow f \geq 2$: range $[2, \infty)$. $|x - 1| \geq 0 \Rightarrow -|x - 1| \leq 0 \Rightarrow g \leq 4$: $(-\infty, 4]$.$\sqrt{x} + 3 \geq 0 \Rightarrow h \geq -2 : [-2, \infty)$. Each answer follows from the extreme value of the inner function.

5. **Let $f(x) = x^2 - 1$, $g(x) = \sqrt{x} + 2$. Find $f + g, f - g$, fg, f/g and each new domain.**
   [Source: Week 2, ISA, task 6]

   Solution. $(f \pm g)(x) = x^2 - 1 \pm \sqrt{x} + 2$; $(fg)(x) = (x^2 - 1)\sqrt{x} + 2$; $(f/g)(x) =$

   $$(f+g)(x) = x^2 + \sqrt{x} + 1; (f-g)(x) = x^2 - \sqrt{x} - 3;$$
   $$(fg)(x) = (x^2 - 1)\sqrt{x} + 2; (\frac{f}{g})(x) = (x^2 - 1)/\sqrt{x} + 2.$$

   $\sqrt{x+2}$. Domains: $f + g, f - g$, fg: $x \geq -2$ (intersection of R and $[-2, \infty)$), $i.e.$ $[-2, \infty)$. For f/g additionally $\sqrt{x} + 2 \neq 0, i.e.$ $x \neq -2$: $(-2, \infty)$.

6. **Let $f(x) = 2x - 3$, $g(x) = x^2 + 1$. Find $(f \circ g)(x)$, $(g \circ f)(x)$, $(f \circ f)(x)$, $(g \circ g)(x)$.**
   [Source: Week 2, ISA, assignment 7]

   Solution. $(f \circ g)(x) = 2x^2 - 1$; $(g \circ f)(x) = 4x^2 - 12x + 10$; $(f \circ f)(x) = 2(2x - 3) - 3 = 4x - 9$; $(g \circ g)(x) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2$. The first two are different, so composition is non-commutative; the last two show that self-composition is perfectly valid.

7. Determine whether each function is even, odd, or of general form: x^4- **2x²;** x^3- **4x;** x^3+x^2**;** $|x|+1$**;** $\frac{x2}{x+1}$**.**

[Source: Week 2, ISA, assignment 8]

Solution. x^4-2x^2: only even powers — even. x^3-4x: only odd powers — odd. x^3+x^2: mixed — general form. $|x|+1$: even

$$x^2/(x+1)$$

(practical exercise 5). $g(-x)=x^2/(-x+1)$: the numerator is even, the denominator changes sign: $g(-x)\neq\pm g(x)$ — general form.

8. Classify: 3x − 7; x^4- **2x** $+1$**;** $^{x+1}x^2+1$**;** 2^x**;** $\sin x$**.** $_{x2+4}$**;**

$$(x^4-2x+1)/(x+1)$$

[Source: Week 2, ISA, assignment 9]

Solution. 3x − 7: linear (polynomial). x^4- 2x $+1$: polynomial. $_{x2+4}$: $\sqrt{x^2}+1$: algebraic (root). 2^x: exponential — transcendental. $\sin x$: trigonometric — transcendental.

9. Linear temperature model from practical exercise 10 — all five parts.

[Source: Week 2, ISA, assignment 10]

Solution. Same model and answers as in practical exercise 7: $T(t)=20+$ 4t; slope = heating rate $4^\circ C/\min$, intercept = initial $20^\circ C$; $T(8)=52^\circ C$; $60^\circ C$ at $t=10\,\min$; domain [0, 15].

10. For $p(x)=x^3-$ **4x and** $r(x)=\frac{x+1}{x-1}$ **determine the domain, intercepts, symmetry, end behavior, points of discontinuity.**

[Source: Week 2, ISA, assignment 12]

Solution. p: as in practical exercise 8 (domain R; intercepts −2, 0, 2; odd; ±∞ at ends; none). r: domain $x\neq 1, i.e.\ (-\infty,1)\cup(1,\infty)$. Intercepts: $r(0)=-1$; numerator zero at $x=-1$. Symmetry: $r(-x)=(-x+1)/(-x-1)=(x-1)/(x+1)\neq\pm r(x)$ — general form. End behavior: $r(x)\to 1$ as $x\to\pm\infty$ (horizontal asymptote $y=1$); $r(x)\to\pm\infty$ as $x\to 1$ (vertical asymptote $x=1$). Undefined only at $x=1$.

11. Piecewise electricity tariff from practical exercise 9 — all five parts.

[Source: Week 2, ISA, assignment 13]

Solution. Same answers as in practical exercise 9: $C(150)=15$, $C(200)=20$, $C(350)=42.5$; continuous broken line graph; slopes — marginal prices; constraint — discontinuity in incentive at the 200 kWh boundary.

12. Convert 30°, 135°, 300° **to radians and** $\frac{\pi}{6}$, $\frac{5\pi}{4}$, $\frac{7\pi}{3}$ **to degrees.**

[Source: Week 2, ISA, assignment 14]

Solution. Multiply by $\pi/180$: 30° $=\pi/6$; 135° $=3\pi/4$; 300° $=5\pi/3$. Multiply by $180/\pi$: $\pi/6=30°$; $5\pi/4=225°$; $7\pi/3=420°$.

13. Is it bijective? *a.* $f(x)=$ **3x** $-5, x\in R$**;** *b.* $g(x)=x^2, x\in R$**;** *c.* $h(x)=x^2, x\geq 0$**;** *d.* $p(x)=\frac{1}{x},\ x\neq 0$**.**

[Source: Week 2, ISA, assignment 15]

Solution. *a.* Yes: non-decreasing line passes horizontal line test. *b.* No: $g(-2)=g(2)=4$. *c.* Yes: squaring is injective on $[0,\infty)$(restriction restores horizontal line test). *d.* Yes: $_x1/1=_x1/2\Rightarrow x_1=x_2$.

14. Find inverses and their domains: $a.$ $f(x) = \mathbf{4x - 7}$**;** $b.$ $g(x) = \frac{x+3}{2}$**;** $c.$ $h(x) = x^2 + 1,\ x \geq 0$**;** $d.$ $p(x) = \frac{2x+3}{x-1},\ x \neq 1$**; verify by composition.**

[Source: Week 2, ISA, assignment 16]

Solution. $a.$ $f^{-1}(x) = (x+7)/4$, domain R (practical exercise 10). $b.$ $y = (x+3)/2 \Rightarrow$

$$y = (x+1)^2 \Rightarrow x = \sqrt{y-1}$$

$x = 2\mathrm{y} - 3$, so $g^{-1}(x) = 2\mathrm{x} - 3$, domain R. $c.$ (positive root, since $x \geq 0$): $h^{-1}(x) = \sqrt{x-1}$, domain $[1, \infty)$. $d.$ $y = (2\mathrm{x}+3)/(x-1) \Rightarrow y(x-1) = 2\mathrm{x}+3 \Rightarrow x(y-2) = y+3 \Rightarrow x = (y+3)/(y-2)$: $p^{-1}(x) = (x+3)/(x-2)$, domain $x \neq 2$. Each satisfies $f(f^{-1}(x)) = x$ by construction.

15. Growth/decay classification for A, B, C, D **(practical exercise 11).**

[Source: Week 2, ISA, assignment 17]

Solution. Same: A growth $(a = 1.08), B$ decay $(a = 0.85), C$ growth $(k = 0.03), D$ decay $(k = -0.4)$.

16. Solve, giving exact answers first: $a.$ $2^{x+1} = 16$**;** $b.$ $5^2x = 125$**;** $c.$ $e^3x = 7$**;** $d.$ $\mathbf{100e}^{-0.2t} = 20$**.**

[Source: Week 2, ISA, assignment 18]

Solution. $a.$ $x + 1 = 4 \Rightarrow x = 3$. $b.$ $2\mathrm{x} = 3 \Rightarrow x = \frac{3}{2}$. $c.$ $3\mathrm{x} = \ln 7 \Rightarrow x = \ln^7 \approx 0.6486$. $3d.$ $e^{-0.2t} = 0.2 \Rightarrow t = -^{ln0.2} = 5 \ln 5 \approx 8.047$. 0.2

17. Battery discharge (exercise 13 from practical session) — all four parts.

[Source: Week 2, ISA, assignment 19]

Solution. Same: $e^{-0.25} \approx 77.9\%$ after $10h$; 50% at $\ln \frac{2}{0}.025 \approx 27.7h$; 10% at $\ln \frac{10}{0}.025 \approx 92.1h$; the exponential never reaches zero in finite time.

18. Bacteria population (exercise 14 from practical session) — all four parts.

[Source: Week 2, ISA, assignment 20]

Solution. Same: $P_0 = 800$; $P(5) \approx 4603$; doubling time $\frac{ln2}{00.35} \approx 1.98h$; purely exponential models fail in the long term (resources, space, waste).

4 Week 3 — Limits (Math I)

4.1 Math I: Formula catalog for week 3

4.1.1 Rates of change and the two founding problems

Limits enter the course through two questions: how fast is a quantity changing (tangent/velocity), and how do function values behave near a point. The average rate over a shrinking interval defines the instantaneous one.

$$m_{\text{sec}} = \frac{f(x) - f(a)}{x - a} \text{ (slope of the secant through } (a, f(a))) \tag{51}$$

$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a} \text{ (average velocity on } [a, t]) \tag{52}$$

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = f'(a) \text{ (instantaneous rate; defined as a limit)} \tag{53}$$

4.1.2 The limit concept, one-sided and infinite limits

The intuitive definition (OpenStax §2.2) and its one-sided variants are the operational heart of the week; the two-sided limit exists exactly when the two one-sided limits agree.

$$\lim_{x\to a} f(x) = L \Leftrightarrow \text{all } f(x)\ (x \neq a) \text{ approach } L \text{ as } x \to a \tag{54}$$

$$\lim_{x\to a^-} f(x) = L, \quad \lim_{x\to a^+} f(x) = L \text{ (left/right limits)} \tag{55}$$

$$\lim_{x\to a} f(x) = L \Leftrightarrow \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L \text{ (existence criterion)} \tag{56}$$

$$\lim_{x\to a^-} f(x) = \pm\infty, \quad \lim_{x\to a^+} f(x) = \pm\infty, \quad \lim_{x\to a} f(x) = \pm\infty \text{ (infinite limits)} \tag{57}$$

$$\lim_{x\to a^-} \frac{1}{(x-a)^n} = +\infty, \quad \lim_{x\to a^+} \frac{1}{(x-a)^n} = +\infty \quad (n \text{ even}) \tag{58}$$

$$\lim_{x\to a^-} \frac{1}{(x-a)^n} = -\infty, \quad \lim_{x\to a^+} \frac{1}{(x-a)^n} = +\infty \quad (n \text{ odd}) : \ x = a \text{ vertical asymptote} \tag{59}$$

The line $x = a$ is a vertical asymptote of f if at least one of the one-sided infinite limits (57) holds there.

4.1.3 The precise definition (§2.5, lecture emphasis)

The $\varepsilon - \delta$ definition converts ”arbitrarily close” into quantified distances; the equivalent absolute-value inequalities are the working tool.

$$\lim_{x\to a} f(x) = L \Leftrightarrow \forall \varepsilon > 0\ \exists \delta > 0 : \ 0 < |x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon \tag{60}$$

$$|f(x) - L| < \varepsilon \Leftrightarrow L - \varepsilon < f(x) < L + \varepsilon \tag{61}$$

$$0 < |x-a| < \delta \Leftrightarrow a - \delta < x < a + \delta \text{ and } x \neq a \tag{62}$$

4.1.4 Two standard limits seen this week

$$\lim_{x\to 0} \frac{\sin x}{x} = 1 \text{ (Table 2.2 of the lecture: values } 0.9983,\ 0.99998,\ 0.9999998\ \ldots) \tag{63}$$

$$\lim_{x\to 0} \sin\frac{1}{x} \text{ does not exist (oscillation)}, \quad \lim_{x\to 0^-} \frac{1}{x} = -\infty, \quad \lim_{x\to 0^+} \frac{1}{x} = +\infty \tag{64}$$

4.2 Math I: ISA for week 3 — questions and complete solutions

1. Study all problems, exercises, and examples presented in the lecture and practical session; be able to answer quickly and without errors.

[Source: Week 3, ISA, assignment 2]

Solution. This is a study instruction, not a calculation; the relevant material is fully solved in the lecture examples, in the practical session block below (exercises 2–8), and in the assigned textbook problems (following items).

2. For $f(x) = x^3$, estimate the slope of the tangent at $x = 1$, using $m_{sec} = f(1+h) - f(1)$

$$m_{sec} = 3 + 3h + h^2$$

from left and right to the same number?

[Source: Week 3, ISA, assignment 3]

Solution. Here $f(1) = 1$ and $f(1+h) = (1+h)^3 = 1+$ 3h + 3h$^2 + h^3$, so $m_{sec} = 3+$ 3h $+h^2$. Values: $h = -0.1 \Rightarrow 2.71$; $-0.01 \Rightarrow 2.9701$; $-0.001 \Rightarrow 2.997$; $0.001 \Rightarrow 3.003$; $0.01 \Rightarrow 3.0301$; $0.1 \Rightarrow 3.31$. Both sides approach 3; indeed $\lim_{h\to 0}(3+$ 3h $+h^2) = 3$ — the exact slope of the tangent line (derivative of 3x^2 at $x = 1$).

3. Solve problems 46–49 and 60–64 from "Chapter 1" (exercise list §2.2) of OpenStax Calculus 1.

[Source: Week 3, ISA, assignment 4]

Solution. These graph reading exercises are written out and fully solved below in §4.4; the numbering in the course sheet refers specifically to this exercise list.

4. A temperature controller outputs $C(T) = 0$ when $T < 50$ and $C(T) = 10$ when $T \geq 50$ (volts). Determine $\lim_{T\to 50-} C(T)$, $\lim_{T\to 50+} C(T)$, $\lim_{T\to 50} C(T)$; interpret as a switching operation.

[Source: Week 3, ISA, assignment 5]

Solution. On the left, the controller reads the "off" branch: $\lim_{T} \to 50 - C(T) = 0$. On the right — the "on": $\lim_{T} \to 50 + C(T) = 10$. Since the one-sided limits are different, the two-sided limit does not exist (56). Interpretation: abrupt switching at $T = 50$ — the output jumps from $0V$ to $10V$, which is exactly how a thermostat controls a heater.

$$s(t) = 2t^2 + 3t(m, s).(1)$$

5. Position of a robot cart: (1) Average speed $s(2+h) - s(2)$

$$s(2+h) - s(2) = (2(2+h)^2 + 3(2+h)) - 14$$

(3) compute $\lim_{h\to 0}$; (4) interpret. h

$$s(3) - 14 = (27 - 14)/(14.72 - 14)$$

[Source: Week 3, ISA, assignment 6]

Solution. $(1)s(2) = 14$. Values: $h = 1$: $= 13$; $h = 0.1$: =

$$s(2+h) - s(2)/h = (14.07(2+h)^2 - 14)/(h) = (14.07(4 + 4h + h^2) - 14)/(h) =$$

$$(56.28 + 56.28h + 14.07h^2 - 14)/(h) = (42.28 + 56.28h + 14.07h^2)/(h)$$
$$42.28/h + 56.28 + 14.07h(9)$$

7.2; $h = 0.01$: $^{14.0702-14} = 7.02$; $h = -0.01$: $^{13.9302-14} = 6.98$. $(2)s(2+h) = 2(2+h)^2 + 3(2+$

$$s(2+h) - s(2) = 2(2+h)^2 + 3(2+h) - (2 \cdot 2^2 + 3 \cdot 2) = 8h + 2h^2, (2)$$

$h) = 8+$ 8h + $2h^2 + 6+$ 3h $= 14+$ 11h + $2h^2$, therefore $^{s(2+h)-s(2)} = ^{11h+2h2} = 11+$ 2h. $(3)hh \lim_{h\to 0}(11+$ 2h$) = 11m/s$. (4) The instantaneous speed of the cart at $t = 2s$ is $11m/s$; the average speeds approach it from both sides.

6. Two systems have outputs $A(t) = (1.08)^t$ and $B(t) = e^{0.03t}$. (1) Compute $\lim_{t\to\infty} A(t)$ and $\lim_{t\to\infty} B(t)$; (2) rewrite $A(t)$ in the form e^{kt}; (3) which grows faster? (4) Compute $\lim_{t\to\infty} A(t)$} for $B(t)$; (5) interpret.

[Source: Week 3, ISA, assignment 7]

Solution. (1) Both bases are greater than 1, so both limits equal $+\infty$. $(2)A(t) = e^{t \cdot ln 1.08}$, and $\ln 1.08 \approx 0.0770$, so $A(t) = e^{0.0770t}$. $(3)A$ grows faster, since its rate constant 0.0770 exceeds 0.03. $(4) \lim_{t\to\infty} B(t)/A(t) = e^{(0.03-0.0770)t} \to 0$: A dominates. (5) Exponential models with different rates diverge exponentially fast.

4.3 Math I: Week 3 practical session (engineering limits) — questions and complete solutions

Course requirements for this session: show mathematical procedure; include units; interpret in engineering terms; specify when the limit does not exist.

1. Solve problems 50–54 from the exercise list §2.2 of the textbook.

[Source: Week 3, practical session, exercise 1]

Solution. Written out and fully solved below in §4.4 (items 50–54).

2. One-sided limits: $f(x) =$ 2x$+1$ when $x < 2, f(x) = 7 - x$ when $x \geq 2$. Find $\lim_{x\to 2} -f, \lim_{x\to 2} +f, \lim_{x\to 2} f, f(2)$; explain whether the two-sided limit exists.

[Source: Week 3, practical session, exercise 2]

Solution. Left branch: $\lim_{x\to 2} -($2x $+1) = 5$. Right branch: $\lim_{x\to 2} +(7-x) = 5$. They are equal, so by $(56) \lim_{x\to 2} f(x) = 5$. Note, $f(2) = 7 - 2 = 5$ as well — the function value equals the limit, so there is no discontinuity in the graph at $x = 2$.

$$V(T) = (2500/T)MB, T \geq 50$$

3. Calibration near a missing measurement: $V(T) = (T^2 - 2500)/(T - 50)$ mV, $T \neq 50(T$ in °$C)$. (1) Why is $V(50)$ undefined? (2) Evaluate $\lim_{T\to 50} V(T)$; (3) how can the limit replace the missing calibration value?

[Source: Week 3, practical session, ex. 3]

Solution. (1) The formula divides by $T - 50$, which equals 0 when $T = 50$; the sensor model simply has no value there. (2) We factor the difference of squares: $T - 2500 = (T - 50)(T + 50)$, so when $T \neq 50, V(T) = T + 50$, and $\lim \frac{2}{T\to 50} V(T) = 100$ mV. (3) The limit describes what the sensor would show at $50^\circ C$; filling the calibration table with 100 mV at this point is justified because the output approaches this value from both sides—it's a "removable" behavior, not a jump.

4. **Switching system:** $V(t) = \mathbf{2t} + 1$ **when** $t < 4$ **and** $V(t) = 12 - t$ **when** $t \geq 4 (V)$**. Determine the one-sided limits, the two-sided limit, and** $V(4)$**; is there** a **sharp voltage change at** $t = 4$**?**

   [Source: Week 3, practical session, ex. 4]

   Solution. From the left: $\lim_{t\to 4} -(2t+1) = 9V$. From the right: $\lim_{t\to 4} +(12-t) = 8V$. $V(4) = 12-4 = 8V$. The one-sided limits are different, so $\lim_{t\to 4} V(t)$ does not exist: at $t = 4$, the voltage drops sharply from $9V$ to $8V$. Engineering significance: $a 1V$ step can excite transients in the controlled device; when steps are harmful, input filters or soft-start logic are used.

5. **Stress-strain:** $\sigma(\varepsilon) = 200000\varepsilon$ **when** $\varepsilon < 0.002$ **and** $\sigma(\varepsilon) = 400 + 10000(\varepsilon - 0.002)$ **when** $\varepsilon \geq 0.002 (\sigma$ **in MPa; Hooke's region, then hardening). Find the one-sided limits at** $\varepsilon = 0.002$**; does the two-sided limit exist? Does the model predict** a **jump? Sketch the graph near the transition.**

   [Source: Week 3, practical session, ex. 5]

   Solution. Elastic side: $\lim_{\varepsilon\to 0.002} -200000\varepsilon = 200000(0.002) = 400$ MPa.

$$\lim_{\varepsilon\to 0.002} -\sigma(\varepsilon) = 200000(0.002) = 400\text{MPa};$$
$$\lim_{\varepsilon\to 0.002} +\sigma(\varepsilon) = 400 + 10000(0.002 - 0.002) =$$

$$400\text{MPa}; \lim_{\varepsilon\to 0.002} \sigma(\varepsilon) = 400\text{MPa}(\text{exists}).$$

   Plastic side: $400 + 10000(\varepsilon - 0.002) = 400$ MPa. Both equal 400 MPa, so the two-sided limit exists and equals 400 MPa; there is no jump—the piecewise model is continuous at the yield point, only the slope changes (from 200,000 to 10,000 MPa per unit strain). The graph is two straight line segments meeting at (0.002, 400) with a distinct corner: elastic loading, then a much shallower plastic branch.

6. **Long-term response of an** RC **circuit:** $V(t) = V_s(1 - e^{-t/(RC)})$ **with** $V_s = 12V$, $R = 10000\Omega$, $C = 100$ **μ**F**. (1) Calculate** $\lim_{t\to 0} +V(t)$**;** $(2) \lim_{t\to\infty} V(t)$**; (3) interpret both physically.**

   [Source: Week 3, practical session, ex. 6]

   Solution. (1) At $t = 0^+$, the exponential equals $e^0 = 1$, so $V(0^+) = 12(1-1) = 0 (2)$ As $t \to \infty, e^{-t/(RC)} \to 0, V$ — initially uncharged capacitor. so $V \to V_s = 12V$. (3) The capacitor charges from $0V$ toward the supply voltage of $12V$, approaching it asymptotically; the time constant $\tau = RC = 10^4 \cdot 10^{-4} = 1s$ sets the pace (about 63% charge after $1s, 99\%$ after 5τ). The limiting language "$V \to V_s$ as $t \to \infty$" is precisely how asymptotic charging is formulated.

7. **Terminal velocity:** $v(t) = v_T(1 - e^{-kt})$, $v_T > 0$, $k > 0$**.** $(1) v(0)$**?** $(2) \lim_{t\to\infty} v(t)$**? (3) Why is this model more realistic than unlimited acceleration?**

   [Source: Week 3, practical session, ex. 7]

   Solution. $(1) v(0) = v_T(1-1) = 0$: the object starts from rest. $(2) e^{-kt} \to 0$, so $v \to v_T$: the speed approaches the terminal value. (3) Air resistance increases with speed until it balances gravity; after that, the net force is zero, and speed can no longer increase—a falling body indeed approaches a finite limit, not accelerating forever.

8. **Data transfer rate:** $R(n) =^{100} n$ **Mbps. (1) Calculate** $n+20 R(10)$, $R(100)$, $R(1000)$**;** $(2) \lim_{n\to\infty} R(n)$**; (3) why does increasing** n **eventually give diminishing improvements? (4) maximum theoretical speed?**

[Source: Week 3, practical session, ex. 8]

Solution. (1)$R(10) \approx 33.3$, $R(100) \approx 83.3$, $R(1000) \approx 98.0$ Mbps. (2) Dividing by n: $R(n) = \frac{100}{1+20/n} \to 100$ Mbps. (3) Each additional unit of resource is divided by an increasingly large base, so the marginal benefit decreases — diminishing returns. (4) The horizontal asymptote 100 Mbps is the ceiling of the model: values approach it but do not exceed it.

4.4 Math I: assigned textbook exercises (OpenStax §2.2, exercises 46–54 and 60–64)

These graph reading exercises use four figures printed with the exercise set; the required values are read from the graphs as in the classroom. First the statements, then the solutions.

1. **From the first graph $y = f(x)$, solve:** (46) $\lim_{x\to 10} -f(x) = 0$**;** (47) $\lim_{x\to -2} +f(x) = 3$**;** (48) $\lim_{x\to -8} f(x) = f(-8)$**;** (49) $\lim_{x\to 6} f(x) = 5$ **— true or false?**
   [Source: OpenStax §2.2, ex. 46–49]

   Solution. (46) True: as $x \to 10$ from the left, the curve descends to the filled point (10, 0), so the left limit is 0. (47) True: slightly to the right of -2, the graph is at height 3 (filled point $(-2, 3)$). (48) False: near $x = -8$, the curve approaches an open circle at height -6, so $\lim_{x\to -8} f(x) = -6$, whereas $f(-8) = -3$ (filled point); the limit and value may not coincide. (49) False: from the left, the curve descends to an open circle (6, 2), so $\lim_{x\to 6} - = 2$, whereas the right branch emerges from the filled point (6, 5), so $\lim_{x\to 6} + = 5$; the one-sided limits are different ⇒ the two-sided limit does not exist, so it cannot equal 5.

2. **For the piecewise linear graph (line $y = x + 1$ to the open point (1, 2); filled point (1, 1); then line $y = 3 - x$), find** (50) $\lim_{x\to 1} -f$**;** (51) $\lim_{x\to 1} +f$**;** (52) $\lim_{x\to 1} f$**;** (53) $\lim_{x\to 2} f$**;** (54)$f(1)$**.**
   [Source: OpenStax §2.2, ex. 50–54]

   Solution. (50) The left branch $y = x + 1 \to 2$. (51) The right branch $y = 3 - x \to 2$. (52) Both sides equal 2, so $\lim_{x\to 1} f(x) = 2$ — even though $f(1) = 1$; the limit ignores the point itself. (53) At $x = 2$, only the right branch exists: $3 - 2 = 1$. (54)$f(1) = 1$ (filled point).

3. **For the third graph (left: line descending to the open point (0, 1); right: curve rising from the filled point (0, −4) and crossing zero near $x = 2$), find** (55) $\lim_{x\to 0} -f$**;** (56) $\lim_{x\to 0} +f$**;** (57) $\lim_{x\to 0} f$**;** (58) $\lim_{x\to 2} f$**.**
   [Source: OpenStax §2.2, ex. 55–58]

   Solution. (55) 1. (56) -4. (57)$1 \neq -4$, so the two-sided limit does not exist (56). (58) The right branch crosses the axis at $x = 2$, so the limit is 0.

4. **For the fourth graph (three pieces: rising line to the open point (−2, 0); descending segment from the filled point (−2, 2) through the origin to (2, −2); rising line from the open point (2, 2)), find** (59) $\lim_{x\to -2} -f$**;** (60) $\lim_{x\to -2} +f$**;** (61) $\lim_{x\to -2} f$**;** (62) $\lim_{x\to 2} -f$**;** (63) $\lim_{x\to 2} +f$**;** (64) $\lim_{x\to 2} f$**.**
   [Source: OpenStax §2.2, ex. 59–64]

   Solution. (59) 0. (60) 2. (61)$0 \neq 2$: does not exist. (62) -2. (63) 2. (64)$-2 \neq 2$: does not exist. Both junctions are actual jumps: no averaging can fix unequal one-sided limits.

5 Week 4 — Limits II: limit laws, standard limits, indeterminacies, asymptotics, Landau symbols (Math I)

Week 4 is the key week before Test 1 review: it turns the limit apparatus of week 3 into a working toolkit. Required reading is OpenStax §2.3 (Limit Laws), and week 4's notes by Prof. Neophytides summarize general limit properties, standard limits, indeterminacies, asymptotic equivalence, and Landau symbols O and o. This section lists each equation from both sources, then solves each exercise in the §2.3 set (exercises 83–130) and each worked question from the notes.

5.1 Formula catalog for week 4

5.1.1 General properties of limits (Week 4 Notes)

$$\lim_{x\to a} f(x) = L: \ f(x) \text{ can be made arbitrarily close to } L \text{ by taking } x \text{ close to } a \ (x \neq a) \tag{78}$$

$$\lim_{x\to a} f(x) = L \Leftrightarrow \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L \text{ (two-sided criterion)} \tag{79}$$

a limit, when it exists, is unique.

5.1.2 Limit laws (OpenStax Theorem 2.5)

If $\lim x \to a f(x) = L$ and $\lim x \to a g(x) = M$ (both existing), and c is a constant, then:

$$\text{Sum: } \lim_{x\to a}[f(x)+g(x)] = L+M; \quad \text{Difference: } \lim_{x\to a}[f(x)-g(x)] = L-M \tag{80}$$

$$\text{Constant multiple: } \lim_{x\to a} c\,f(x) = cL; \quad \text{Product: } \lim_{x\to a}[f(x)g(x)] = LM \tag{81}$$

$$\text{Quotient: } \lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M} \ (M \neq 0); \quad \text{Power: } \lim_{x\to a}[f(x)]^n = L^n \ (n \in \mathbb{N}) \tag{82}$$

$$\text{Root: } \lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{L} \ (n \text{ odd: any } L;\ n \text{ even: } L \geq 0) \tag{83}$$

$$\text{Basic: } \lim_{x\to a} x = a, \quad \lim_{x\to a} c = c \tag{84}$$

$$\text{Polynomials/rational functions: } \lim_{x\to a} p(x) = p(a), \quad \lim_{x\to a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)} \ (q(a) \neq 0) \tag{85}$$

If $f(x) = g(x)$ for all $x \neq a$ near a, then $\lim x \to a f(x) = \lim x \to a g(x)$ — the licence to cancel factors before substituting.

5.1.3 The Squeeze (Pinching) Theorem (OpenStax Theorem 2.7)

$$f(x) \leq g(x) \leq h(x) \text{ near } a \ (x \neq a), \quad \lim_{x\to a} f(x) = L = \lim_{x\to a} h(x) \Rightarrow \lim_{x\to a} g(x) = L \tag{86}$$

5.1.4 Standard limits and the derived trigonometric limits (Week 4 Notes)

$$\lim_{x\to 0}\frac{\sin x}{x}=1,\quad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2},\quad \lim_{x\to 0}\frac{e^x-1}{x}=1 \tag{87}$$

$$\lim_{x\to 0}\frac{\ln(1+x)}{x}=1,\quad \lim_{x\to 0}(1+x)^{1/x}=e,\quad \lim_{x\to\infty}\frac{x^n}{e^x}=0\ (n\in\mathbb{N}) \tag{88}$$

Angles in the trigonometric limits must be in radians. Adapting (87) by substitution, *e.g.*:

$$\lim_{x\to 0}\frac{\sin 5x}{x}=5\lim_{x\to 0}\frac{\sin 5x}{5x}=5,\quad \lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1\Rightarrow\lim_{\theta\to 0}\frac{1-\cos\theta}{\theta}=0\ \text{(via conjugate)} \tag{89}$$

5.1.5 Indeterminate forms (Week 4 Notes)

$$\frac{0}{0},\ \frac{\infty}{\infty},\ 0\cdot\infty,\ \infty-\infty,\ 0^0,\ 1^\infty,\ \infty^0 \tag{90}$$

The word ”indeterminate” does not mean that the limit fails to exist; it means that more analysis is required. The same initial form can resolve differently:

$$\lim_{x\to 0}\frac{x}{x}=1,\quad \lim_{x\to 0}\frac{x}{x^2}=0,\quad \lim_{x\to 0}\frac{x^2}{x}\ \text{does not exist (finite)} \tag{91}$$

while $\lim x\to 1(x^2-1)/(x-1)=2$ after factoring. Resolution tools: factoring, rationalisation, standard limits, asymptotic equivalence, change of variable, algebraic rearrangement, the Squeeze Theorem, Taylor expansions, and (later) L’Hôpital’s rule.

5.1.6 Asymptotic equivalence (Week 4 Notes)

$$f(x)\sim g(x)\ (x\to a)\Leftrightarrow\lim_{x\to a}\frac{f(x)}{g(x)}=1 \tag{92}$$

$$f(x)\sim g(x)\Leftrightarrow f(x)=g(x)\big(1+\varepsilon(x)\big),\ \varepsilon(x)\to 0 \tag{93}$$

$$f\sim g\Leftrightarrow f=g+o(g) \tag{94}$$

$$\text{as } x\to 0:\ \sin x\sim x,\ \tan x\sim x,\ 1-\cos x\sim\frac{x^2}{2},\ e^x-1\sim x \tag{95}$$

$$\ln(1+x)\sim x,\quad (1+x)^\alpha-1\sim\alpha x\ (\alpha\ \text{constant}) \tag{96}$$

Numerically, $\sin(0.01)/0.01\approx 0.999983$ illustrates ”close, not equal”. At infinity the highest-degree term dominates:

$$3x^2+5x-1\sim 3x^2\ (x\to\infty),\quad \frac{2x^3+x}{5x^2-1}\sim\frac{2x^3}{5x^2}=\frac{2}{5}x\ (x\to\infty) \tag{97}$$

Equivalences may be multiplied and divided, not generally added.

5.1.7 Landau symbols: big$-O$ and little$-o$ (Week 4 Notes)

$$f(x) = O\big(g(x)\big)\ (x \to a) \Leftrightarrow \frac{|f(x)|}{|g(x)|} \text{ remains bounded near } a \tag{98}$$

$$f(x) = o\big(g(x)\big)\ (x \to a) \Leftrightarrow \lim_{x \to a} \frac{f(x)}{g(x)} = 0 \tag{99}$$

$$f = o(g) \Rightarrow f = O(g); \quad f = O(g) \not\Rightarrow f = o(g) \tag{100}$$

$$x^2 = o(x),\ x^3 = o(x^2)\ (x \to 0) \tag{101}$$

$$x = o(x^2),\ x^{10} = o(e^x)\ (x \to \infty) \tag{102}$$

$$3x = O(x) \text{ but } 3x \neq o(x)\ (x \to 0) \text{ since } \frac{3x}{x} = 3 \tag{103}$$

$$\sin x = x + o(x)\ (x \to 0); \quad e^x = 1 + x + o(x); \quad e^x = 1 + x + \frac{x^2}{2} + o(x^2) \tag{104}$$

$$2x^2 + 3x + 1 = 2x^2 + O(x)\ (x \to \infty) \tag{105}$$

In words: big$-O$ means f has at most the order of magnitude of g; little$-o$ means f becomes negligible compared with g. Caution: $f = O(g)$ is not an equation in the ordinary sense — $O(g)$ denotes a whole class of functions.

5.2 OpenStax §2.3, check problems 2.11–2.20 — questions and complete solutions

1. Using the limit laws, compute $\lim_{x\to -3}(2x-1)\sqrt{x}+4$, indicating the law at each step.

[Source: OpenStax §2.3, check 2.11]

Solution. Both limits of the factors exist, so we apply the product law: $\lim(2x-1) = 2(-3)-1 = -7$(sum, difference, basic limits); $\lim \sqrt{x}+4 = \sqrt{-3}+4 = 1$ (root law). Product: $-7 \cdot 1 = -7$.

2. Compute $\lim_{x\to -2}(3x^3 - 2x + 7)$.

[Source: Check 2.12]

Solution. Direct substitution is justified (85): $3(-8) - 2(-2) + 7 = -24 + 4 + 7 = -13$.

3. Compute $\lim_{x\to -3}(x^2 + 4x + 3)/(x^2 - 9)$.

[Source: Check 2.13]

Solution. 0/0 form. Factor both: $(x^2+4x+3)/(x^2-9) = ((x+1)(x+3))/((x+3)(x-3)) = (x+1)/(x-3)$ when $x \neq -3$. Limit: $(-3+1)/(-3-3) = (-2)/(-6) = \frac{1}{3}$.

4. Compute $\lim_{x\to 5}(\sqrt{x-1}-2)/(x-5)$.

[Source: Check 2.14]

Solution. 0/0 form. Multiply by the conjugate $\sqrt{x-1}+2$:

$$(\sqrt{x-1}-2)/(x-5) = (x-1-4)/((x-5)(\sqrt{x-1}+2))$$
$$1/(\sqrt{x-1}+2)(x \neq 5)$$

so the limit is $1/(\sqrt{5-1}+2) = \frac{1}{4}$.

5. Compute $\lim_{x\to -3}(1/(x+2)+1)/(x+3)$**.**
[Source: Check 2.15]

Solution. Simplify the complex fraction: $1/(x+2)+1 = (x+3)/(x+2)$, so the expression equals $(x+3)/((x+2)(x+3)) = 1/(x+2)$ when $x \neq -3$. Limit: $1/(-3+2) = -1$.

6. Compute $\lim_{x\to 3}(1/(x-3) - 4/(x^2 -$ **2x − 3)).**

$$1/(x-3) - 1/(x+1)$$

[Source: Check 2.16]

Solution. Factor $x^2 -$ 2x $-3 = (x-3)(x+1)$ and simplify:

$$1/(x-3) - 4/((x-3)(x+1)) = ((x+1)-4)/((x-3)(x+1))$$
$$(x-3)/((x-3)(x+1)) = 1/(x+1)$$

$$(x \neq 3),$$

so the limit is 1/4.

7. Piecewise check: $f(x) = x^2 - x$ **for** $x \leq 3$ **and** $f(x) = x+4$ **for** $x > 3$**; compute the one-sided limits at the junction.**
[Source: Check 2.17]

Solution. On the left, the parabolic branch gives $\lim_{x\to 3-} f(x) = 9$; on the right, the straight line gives $\lim_{x\to 3} +(x+4) = 7$. Not equal, so $\lim_{x\to 3} f(x)$ does not exist.

8. Compute $\lim_{x\to 1}(x+2)/(x-1)^2$ **(method of example 2.23).**
[Source: Check 2.18]

Solution. Numerator $\to 3$ (nonzero), denominator $\to 0^+$ from both sides (square), so the ratio → +∞: infinite limit, and $x = 1$ is a vertical asymptote.

9. Using the squeeze theorem, compute $\lim_{x\to 0} x^2 \sin\frac{1}{x}$**.**
[Source: Check 2.19]

Solution. Since $-1 \leq \sin\frac{1}{x} \leq 1$, multiplying by $x^2 \geq 0$ gives $-x^2 \leq x^2 \sin 1_x \leq x^2$. Both bounds $\to 0$; by (86) the limit is 0.

10. Compute $\lim \frac{1-cos\theta}{\theta\to 0 sin\theta}$**.**
[Source: Check 2.20]

Solution. Divide numerator and denominator by θ (legal since within the limit $\theta \neq 0$): $1 - \cos\theta 1 - \cos\theta - \to 0\theta$

$$(1-\cos\theta)/(a\sin\theta) = 0, 1$$

$\sin\theta \sin\theta\theta$ using $^{1-cos\theta} \to 0$ (example 2.25) and $^{sin\theta} \to 1$ (standard limit). $\theta\theta$

5.3 OpenStax §2.3, exercises 83–130 — questions and complete solutions

5.3.1 Using limit laws (83–86)

1. $\lim_{x\to 0}(\mathbf{4x}^2 - \mathbf{2x} + 3)$**.**
[Source: §2.3, ex. 83]

Solution. Substitution by (85): $4(0)^2 - 2(0) + 3 = 3$.

2. $\lim_{x\to 1}(x^3 + \mathbf{3x}^2 + 5)/(4 - \mathbf{7x})$**.**
[Source: §2.3, ex. 84]

Solution. Denominator at 1 is $-3 \neq 0$, so quotient law and substitution give $= \frac{1+3+5}{-}9/3 = -3$. $-3\surd$

$$\lim_{x\to 9} \sqrt{x^2 - 6x + 3}.$$

3. $\lim_{x\to -2}$
[Source: §2.3, ex. 85]

Solution. The radicand is continuous at -2 and positive there; root law: $\sqrt{4 + 12 + 3} = \sqrt{19}$.

4. $\lim_{x\to -1}(\mathbf{9x} + 1)^2$**.**
[Source: §2.3, exercise 86]

Solution. Power law, then substitution: $(9(-1) + 1)^2 = (-8)^2 = 64$.

5.3.2 Direct Substitution (87–92)

1. $\lim_{x\to 7} x^2$**.**
[Source: §2.3, exercise 87]

Solution. 49.

2. $\lim_{x\to -2}(\mathbf{4x}^2 - 1)$**.**
[Source: §2.3, exercise 88]

Solution. $4(4) - 1 = 15$.

3. $\lim_{x\to 0} 1/(1 + \sin x)$**.**
[Source: §2.3, exercise 89]

Solution. Denominator $1 + \sin 0 = 1 \neq 0$: quotient law gives $\frac{1}{1} = 1$.

4. $\lim_{x\to 2} e^{2x - x2}$**.**
[Source: §2.3, exercise 90]

Solution. Exponent: $2(2) - 4 = 0$; $e^0 = 1$.

5. $\lim_{x\to 1}(2 + \mathbf{7x})/(x + 6)$**.**
[Source: §2.3, exercise 91]

Solution. $\frac{2+7}{1+6} = 9_7$**. 6.** $\lim_{x\to 3} \ln e^3 x$**.**
[Source: §2.3, exercise 92]

Solution. $\ln e^3 x = 3$x identically (logarithm and exponential are inverses), so the limit is $3 \cdot 3 = 9$.

5.3.3 Indeterminate Form 0/0(93–102)

1. $\lim_{x\to4}(x^2-16)/(x-4)$**.**
[Source: §2.3, exercise 93]
Solution. Form 0/0. Factor: $^{(x-4)(x+4)} = x+4$ when $x \neq 4$; limit $= 8$. $x-4$

2. $\lim_{x\to2}(x-2)/(x^2-$ **2x).**
[Source: §2.3, exercise 94]
Solution. Factor denominator: $\frac{x-2}{x(x-2)} = 1_x$ when $x \neq 2$; limit $= 1_2$.

3. $\lim_{x\to6}($**3x − 18)/(2x − 12).**
[Source: §2.3, exercise 95]
Solution.$\frac{3(x-6)}{2(x-6)} = 3_2$ when $x \neq 6$; limit $= 3_2$.

$$\lim_{h\to0}((1+h)^2-1)/h$$

4. $\lim_{h\to0}$**.** h
[Source: §2.3, exercise 96]
Solution.$^{1+2h+h2-1} = {}^{h(2+h)} = 2+h$; limit $= 2$ (this is the derivative of x^2 at 1). hh

5. $\lim_{t\to9}(t-9)/(\sqrt{t}-3)$**.**
[Source: §2.3, exercise 97]
Solution. Form 0/0. Multiply by conjugate of denominator: √

$$\lim_{t\to9}(t-9)/(\sqrt{t}-3) = \lim_{t\to9}(t-9)(\sqrt{t}+3)/((\sqrt{t}-3)(\sqrt{t}+3))$$
$$\lim_{t\to9}(t-9)(\sqrt{t}+3)/(t-9) = \lim_{t\to9}$$

$$(\sqrt{t}+3) = 6(t \neq 9)$$

so the limit is $9+3=6$.

6. Find $\lim_{h\to0}(1/(a+h)-1/a)/h$**, where** a **is** a **nonzero real constant.**
[Source: §2.3, exercise 98]
Solution. Simplify numerator: $1/a-1/(a+h) = (a-(a+h))/(a(a+h)) = -h/(a(a+h))$. Divide by h: $-1/(a(a+h))$ for $h \neq 0$.
Limit: $-1/a^2$ (derivative of $1/x$ at a).

7. $\lim\frac{sin\theta}{\theta\to\pi tan\theta}$**.**
[Source: §2.3, exercise 99]
Solution. When $\theta \neq \pi$ near π: $\sin\theta/\tan\theta = \sin\theta\cdot\cos\theta/\sin\theta = \cos\theta$ (cancellation is valid since $\sin\theta \neq 0$ there). Limit: $\cos\pi = -1$.

8. $\lim_{x\to1}(x^3-1)/(x^2-1)$**.**
[Source: §2.3, exercise 100]
Solution. Factor both: $^{(x-1)(x2+x+1)} = {}^{x2+x+1}$; limit $= \frac{3}{2}$.

$$(x-1)(x+1)/(x+1)$$

9. $\lim_{x\to 1/2}(\mathbf{2x^2+\ 3x\ -\ 2)/(2x\ -\ 1).}$

[Source: §2.3, exercise 101]

Solution. $2x^2+\ 3x\ -2 = (2x\ -1)(x+2)$; cancel to $x+2$; limit $= \frac{1}{2}+2 = 5_2$.

$$\sqrt{x+4}-1$$

10. $\lim_{x\to -3}(\sqrt{x+4}-1)/(x+3)$**.**

$$\sqrt{x+4+1}$$

[Source: §2.3, exercise 102]

Solution. Conjugate: multiply by $\sqrt{x+4+1}$:

$$((x+4)-1)/\sqrt{x+4+1} = 1/((x+3)(x+4+1)), (x\neq -3);$$

limit $= \frac{1}{1+1} = 1_2$.

5.3.4 Forms $K/0$: One-sided Infinite Limits (103–106)

1. $\lim_{x\to -2-}(\mathbf{2x^2+\ 7x}\ -4)/(x^2+x-2)$**.**

$$(x+2)/(x-2)$$

[Source: §2.3, exercise 103]

Solution. Factor: numerator = $(2x-1)(x+4)$, denominator $= (x+2)(x-1)$. At $x=-2$, numerator $= (-5)(2) = -10 \neq 0$, while denominator $\to 0$: form $K/0$. Sign near -2: take $x=-2.1$: $(x+2)<0, (x-1)<0$, so denominator is positive; numerator $\approx -10<0$. Ratio $\to -\infty$.

2. $\lim_{x\to -2+}(\mathbf{2x^2+\ 7x}\ -4)/(x^2+x-2)$**.**

$$(x+2)/(x-2)$$

[Source: §2.3, exercise 104]

Solution. Same numerator; now $(x+2)>0, (x-1)<0$: denominator is negative. Ratio $\to +\infty$.

3. $\lim_{x\to 1-}(\mathbf{2x^2+\ 7x}\ -4)/(x^2+x-2)$**.**

$$(x+2)/(x-2)$$

[Source: §2.3, exercise 105]

Solution. Numerator at 1: $(1)(5) = 5 > 0$. When $x\to 1^-$: $(x+2)>0, (x-1)<0$, denominator is negative: ratio $\to -\infty$.

4. $\lim_{x\to 1+}(\mathbf{2x^2+\ 7x}\ -4)/(x^2+x-2)$**.**

$$(x+2)/(x-2)$$

[Source: §2.3, ex. 106]

Solution. The denominator is positive to the right of 1: the ratio $\to +\infty$. (The two-sided limits at -2 and at 1 both do not exist; $x = -2$ and $x = 1$ are vertical asymptotes.)

5.3.5 Limit Laws with Given Limits (107–114)

Throughout: $\lim_{x\to 6} f(x) = 4$, $\lim_{x\to 6} g(x) = 9$, $\lim_{x\to 6} h(x) = 6$.

1. $\lim_{x\to 6}$ **2f** $(x)g(x)$**.**
[Source: §2.3, ex. 107]
Solution. Product, then constant factor: $2 \cdot 4 \cdot 9 = 72$.

$$: g(x) - 1$$

2. $\lim_{x\to 6}(g(x) - 1)/f(x)$**.**
[Source: §2.3, ex. 108]
Solution. Law of quotients: $^{9-1} = 2$.

$$\lim_{x\to 6} f(x).$$

3. $\lim_{x\to 6} f(x) + \frac{1}{3}g(x)$**.**
[Source: §2.3, ex. 109]
Solution. Sum and constant factor: $4 + \frac{9}{3} = 7$.

4. $\lim_{x\to 6} h(x)^3$**.**
[Source: §2.3, ex. 110]
Solution. Law of powers: $6^3 = 216$.

5. $\lim_{x\to 6} \sqrt{g(x) - f(x)}$**.**
[Source: §2.3, ex. 111]
Solution. Difference, then root: $\sqrt{9} - 4 = \sqrt{5}$.

6. $\lim_{x\to 6} x \cdot h(x)$**.**
[Source: §2.3, ex. 112]
Solution. Law of product with $\lim x = 6$: $6 \cdot 6 = 36$.

7. $\lim_{x\to 6}(x + 1) \cdot f(x)$**.**
[Source: §2.3, ex. 113]
Solution. $(6 + 1) \cdot 4 = 28$.

8. $\lim_{x\to 6} f(x)g(x) - h(x)$**.**
[Source: §2.3, ex. 114]
Solution. $4 \cdot 9 - 6 = 30$.

5.3.6 Piecewise Functions and Limits from Graphs (115–125)

1. $f(x) = (x^2,\ x \le 3; x + 4,\ x > 3)$**: find** a**.** $\lim_{x\to 3-} f, b$**.** $\lim_{x\to 3+} f$**.**
[Source: §2.3, ex. 115]
Solution. a. Parabolic branch: 9. b. Linear branch: 7. (Two-sided limit does NOT exist.)

2. $g(x) = (x^3 - 1,\ x \le 0; 1,\ x > 0)$**: find** a**.** $\lim_{x\to 0-} g, b$**.** $\lim_{x\to 0+} g$**.**

$$0^3 - 1 = -1.$$

[Source: §2.3, ex. 116]

Solution. a. Cubic branch: $0^3 - 1 = -1$. b. Constant branch: 1. (Two-sided does NOT exist.)

3. $h(x) = (x^2 -$ **2x** $+1, x < 2$**;** $3 - x, x \ge 2)$**: find** a**.** $\lim_{x\to 2-} h, b$**.** $\lim_{x\to 2+} h$**.**
[Source: §2.3, ex. 117]

Solution. a. $4 - 4 + 1 = 1$. b. $3 - 2 = 1$. Both sides agree: $\lim_{x\to 2} h(x) = 1 = h(2)$ — continuous there, despite the piecewise definition.

4. Let f **and** g **be two depicted functions (V-shaped** $f(x) = |x + 2|$**; piecewise** g **with linear left branch** $g(x) = -x - 6$ **and curvilinear right branch), calculate** $\lim_{x\to -3} +(f(x) + g(x))$**.**
[Source: §2.3, ex. 118]

Solution. From the graph: $f(-3) = | - 3 + 2| = 1$ and on the curvilinear branch $g(-3^+) = 1$. Law of sum: $1 + 1 = 2$.

$$f(-3) - 3g(-3^-) = 1 - 3(-1) = 4.$$

[Source: §2.3, ex. 119]

Solution. Left linear branch gives $g(-3^-) = -(-3) - 6 = -3$... reading the junction from the graph, the value of the left branch at -3 is -1; we take the graph-value $g(-3^-) = -1$: $f(-3) -$ 3g$(-3^-) = 1 - 3(-1) = 4$.

6. $\lim_{x\to 0} f(x)g(x)/3$**.**
[Source: §2.3, ex. 120]

Solution. $f(0) = |2| = 2$; $g(0) = -2$(the curve passes through (0, −2)). Product and quotient: $2 \cdot (-2)/3 = -\frac{4}{3}$.

$$7. \lim_{x\to 3}(2 + g(x))$$

7. $\lim_{x\to -3}(2 + g(x))/f(x)$**.**

$$(2 + g)/(-3 - 2) + (-1)/(2 + 1)$$

[Source: §2.3, ex. 121]

Solution. Left: $\lim_{x\to -3} f(x)\} = 1$; right: $\lim_{x\to -3} f(x)\} = 3$. One-sided

$$f(-3) = \frac{1}{2}$$

limits are different, so two-sided limit does not exist. $_2$.

8. $\lim_{x\to 1} f(x)^2$**.**
[Source: §2.3, ex. 122]

Solution. $f(1) = |3| = 3$; law of powers: 9.

$$P_3 = (1)/(3); \text{canonical form} : 3.$$

9. $\lim_{x\to 1} \sqrt[3]{f(x)-g(x)}$**.**
[Source: §2.3, ex. 123]

Solution. From the graph, $g(1)=-5$ and $f(1)=3$: difference 8; cube root: $\sqrt[3]{8}=2$.

10. $\lim_{x\to -7} xg(x)$**.**
[Source: §2.3, ex. 124]

Solution. At $x=-7$ the linear branch gives $g(-7)=-(-7)-6=1$; product: $(-7)(1)=-7$.

11. $\lim_{x\to -9}[xf(x)+2g(x)]$**.**
[Source: §2.3, ex. 125]

Solution. $f(-9)=|-7|=7$; $g(-9)=9-6=3$; $(-9)(7)+2(3)=-63+6=-57$.

5.3.7 Squeeze Theorem and Applications (126–130)

T True or false: if 2x$-1\le g(x)\le x^2-$2x$+3$ near $x=2$, then $\lim_{x\to 2} g(x)=0$.
[Source: §2.3, ex. 126]

Solution. Incorrect. Both bounds at $x=2$ equal $2(2)-1=3=4-4+3$, so by the squeeze theorem $\lim_{x\to 2} g(x)=3$, not 0. (The theorem gives exactly the common limit of the bounds.)T $\lim_{\theta\to 0}\theta^2\cos\frac{1}{\theta}$.
[Source: §2.3, ex. 127]

Solution. $-1\le\cos\frac{1}{\theta}\le 1$ gives $-\theta^2\le\theta^2\cos 1_\theta\le\theta^2$; both bounds $\to 0$; squeeze: limit $=0$. (0, x rational

1. $\lim_{x\to 0} f(x)$**, where** $f(x)=x$ **irrational.**

$$\lim_{x\to 0} f(x)=0$$

[Source: §2.3, ex. 128]

Solution. For all x: $0\le f(x)\le x^2$ (rational x gives 0; irrational x gives $x^2\ge 0$). Since $0\to 0$ and $x^2\to 0$, the squeeze theorem forces $\lim_{x\to 0} f(x)=0$ — even though f is wildly discontinuous at every point $x\neq 0$.

$$\text{Coulomb}'s\text{law}: E(r)=q/(4\pi\varepsilon_0 r^2)\text{with}q=10^{-19};$$
$$\text{compute}\lim_{r\to 0+} E(r)\text{and interpret; why onthe right?}$$

interpret; why right?
[Source: §2.3, ex. 129]

Solution. The numerator is a positive constant; as $r\to 0^+$ the denominator $\to 0^+$, so $E(r)\to+\infty$: the field magnitude grows without bound as the test point approaches the charge. Only $r\to 0^+$ makes sense, because r is distance and cannot be negative (and $r=0$ is the location of the charge itself). T Density $\rho=m/V$ gives $V(\rho)=\frac{m}{\rho}$; when $m=8$ kg compute $\lim_{\rho\to 0}+V(\rho)$ and explain the physical meaning.
[Source: §2.3, ex. $130]_\rho$

; as $\rho\to 0^+$ volume → +∞:

Solution. $V(\rho)=m/\rho$: an object of fixed mass but vanishing density must occupy infinite volume (the formal statement of an "ideal gas, thinning out"): $V(\rho)=m/\rho\to+\infty$ as $\rho\to 0^+$. The right-hand limit is the only physical option, since density is positive.

5.4 Solved examples from week 4's notes — questions and complete solutions

1. **Show that** $\lim_{x\to1}(x^2-1)/(x-1)$ **has the form 0/0, and compute it; compare with** $\lim_{x\to0}x/x$, $\lim_{x\to0}x^2/x$, $\lim_{x\to0}x/x^2$**.**
   [Source: Week 4 notes, indeterminacies]

   Solution. Substituting $x=1$ gives 0/0 — indeterminacy. Factoring: $(x^2-1)/(x-1)=(x-1)(x+1)/(x-1)=x+1$ when $x\neq1$, so the limit is 2. The three comparisons resolve

   2

   the same form 0/0 in three different outcomes: $=1\to1$; $=x\to0$; =

   $$\lim_{x}(x-1)/(x-1)$$

   →±∞ (no finite two-sided limit exists). This is why 0/0 carries no automatic value: the outcome depends on how the numerator and denominator approach zero.

2. **Using the standard limit** $\lim_{x\to0}\sin x/x=1$**, compute** $\lim_{x\to0}x/\sin$ **5x.**
   [Source: Week 4 notes, standard limits]

   Solution. Let $u=$ 5x (so $u\to0$ as $x\to0$): sin 5x/$x=5\cdot$sin 5x/5x $=5\cdot\sin u/u\to5\cdot1=5$.

3. **Using asymptotic equivalence, compute** $\lim_{x\to0}\sin$ **3x/5x and** $\lim_{x\to0}(1-\cos x)/x$**.**

   $$\sin3x/5x\sim3x/5x$$

   [Source: Week 4 notes, asymptotics]

   Solution. Since sin 3x ~ 3x as $x\to0,$sin 3x/5x ~ 3x/5x $=\frac{3}{5}$, so the limit is 3/5. Since $1-\cos x\sim x^2/2,(1-\cos x)/x^2\sim$ 1/2, so the limit is 1/2.

   $$\lim_{x\to0}(\sin^3x)/(5x)=\frac{1}{5}(16)$$

   the standard limit (87) via equivalence.

4. **Show that 3x**^{2}**+ 5x − 1 ~ 3x**^{2} **as** $x\to\infty$**, and compute** $\lim_{x\to\infty}(\mathbf{2x}^3+x)/(\mathbf{5x}^2-1)$ **via equivalences.**
   [Source: Week 4 notes, equivalence at infinity]

   Solution. (3x^2+ 5x − 1)/3x^2 $=1+5/(3x)-1/(3x^2)\to1$ as $x\to\infty$, therefore 3x^2+ 5x − 1 ~ 3x^2.

   $$\lim_{x\to\infty}(3x^2+5x-1)/(3x^2)=1+0-0=1,\text{equivalence established}:$$

   the highest degree term governs long-term behavior. Then $(2x^3+x)/(5x^2-1)\approx(\frac{2}{5})x$. $\lim_{x\to\infty}(2x^3+x)/(5x^2-1)=(\frac{2}{5})x$, which → +∞; more precisely, the ratio of the original fraction to $(\frac{2}{5})x$ approaches 1.

5. **Show that** $\sin x=O(x)$ **as** $x\to0$**, and that 3x** $=O(x)$**, but 3x** $\neq o(x)$**.**
   [Source: Week 4 Notes, Landau symbols]

   Solution. The inequality $|\sin x|\leq|x|$ holds for all real $x(a$ standard geometric fact), which precisely means

$$|\sin x| \le 3|x|$$

boundedness of $|\sin x|/|x|$: $\sin x = O(x)$. For 3x: $|3x|/|x| = 3$ is bounded, so 3x $= O(x)$; but $3x/x = 3 \to 0$, therefore 3x $\neq o(x)$. This is the most vivid illustration of the difference between O and o: big O allows any bounded ratio, little o requires the ratio to vanish.

6. Verify $x^2 = o(x)$ as $x \to 0$ and $x = o(x^2)$, $x^{10} = o(e^x)$ as $x \to \infty$.

$$x^2 - 2x + x = o(x) \text{near zero.}$$

[Source: Week 4 Notes, Landau symbols]

$o(x^{1/2})$; $x^{10}/e^x \to 0$ by the standard limit $\lim_{x\to\infty} x^n/e^x \to 0(88)$: exponential growth eventually dominates any fixed power.

7. Write $\sin x \sim x$ and $e^x = 1 + x + x^2/2 + o(x^2)$ in terms of little o. Why is $f(x) = O(g(x))$ not an ordinary equation?

[Source: Week 4 Notes, relations]

Solution. By $(94), f \sim g$ if and only if $f = g + o(g)$: thus $\sin x \sim x$ is rewritten as $\sin x = x + o(x)$ — the error is negligible compared to x. For the exponential $e^x = 1 + x + o(x)$ holds because $^{ex-1-x} \to 0, x$ and keeping one more term gives $e^x = 1 + x + ^x 2/2 + o(x^2)$. Finally, $f = O(g)$ is not an equation because $O(g)$ denotes an entire class of functions bounded by constant multiples of g; the notation $2x^2$+3x $+1 = 2x^2 + O(x)$ means precisely that the remainder 3x $+1$ grows no faster than a constant multiple of x.

6 Textbook supplements — additional exercises for required reading with full solutions

6.1 OpenStax §1.2 (Basic Classes of Functions) — selected exercises solved

1. For each pair of points, find the slope of the line through them and say whether the line is increasing, decreasing, horizontal, or vertical: (59) (−2, 4) and (1, 1); (61) (3, 5) and (−1, 2); (63) (2, 3) and (5, 7); (65) (2, 4) and (1, 4); (66) (1, 4) and (1, 0).

[Source: OpenStax §1.2, exercises 59–66]

Solution. Slope $m = (y_2 - y_1)/(x_2 - x_1)$. $(59) m = (-5 - 1)/(2 - (-4)) = -\frac{6}{6} = -1$: decreasing. $(61) m = 4 > 0$: increasing. $(63) m = (7 - 3)/(5 - 2) = \frac{4}{3}$: increasing. $(65) m = (4 - 4)/(1 - 2) = 0$: horizontal. $(66) \Delta x = 0$: vertical, slope undefined.

2. Write the equation in slope-intercept form: (67) slope −6, through (1, 3); (70) slope $\frac{2}{5}, x-$intercept 8; (71) through (2, 1) and (−2, −1); (73) intercepts: x at 5 and y at −3.

[Source: OpenStax §1.2, exercises 67–73]

Solution. $(67) y - 3 = -6(x - 1) \Rightarrow y = -6x + 9$. $(70) x-$intercept at 8 means the point (8, 0): $y - 0 = \frac{2}{5} \cdot (x - 8) \Rightarrow y = \frac{2}{5} \cdot x - \frac{16}{5}$. $(71) m = (-1 - 1)/(-2 - 2) = -2/-4 = \frac{1}{2}$; $y - 1 = \frac{1}{2} \cdot (x - 2) \Rightarrow y = \frac{1}{2} \cdot x$. (73) through (5, 0) and (0, −3): $m = (-3 - 0)/(0 - 5) = \frac{3}{5}$; $y = \frac{3}{5} \cdot x - 3$.

3. **(75–76) For each linear equation, specify the slope m and intercept b: $y =$ 2x − 3; $y = -\frac{1}{7}x + 1$.**
   [Source: OpenStax §1.2, exercises 75–76]

   Solution. $(75) m = 2, b = -3$. $(76) m = -\frac{1}{7}, b = 1$.

4. **(81–82) Rewrite with slope and intercept: 2x + 3y $= 6$; 6x − 5y $+15 = 0$.**
   [Source: OpenStax §1.2, exercises 81–82]

   Solution. $(81) y = 2 - \frac{2}{3}x$: $m = -\frac{2}{3}, b = 2$. $(82) y = 6_5x + 3$: $m = 6_5, b = 3$.

5. **(83–86) For each polynomial: degree; zeros; $y-$intercept; behavior at the ends; parity. a. $f(x) =$ 2x$^2-$ 3x − 5; c. $f(x) = x^3+$ 3x$^2 - x - 3$.**
   [Source: OpenStax §1.2, exercises 83–86]

   Solution. a. Degree 2; 2x$^2-$3x$-5 =$ (2x$-5)(x + 1)$: zeros 5/2 and -1; $y-$intercept -5; branches upward (leading coefficient $2 > 0$), so $f \to +\infty$ at both ends; neither even nor odd (due to the odd term $-$3x and the constant). c. Degree 3; testing $x = -1$: $-1 + 3 + 1 - 3 = 0$, so $(x + 1)$ is a factor: x^3+ 3x$^2 - x - 3 = (x + 1)(x^2 - 1) =$ $(x+1)(x-1)(x+1) = (x-1)(x+1)^2$: zeros 1 (simple) and -1 (double); $y-$intercept -3; cubic with positive leading coefficient: $f \to +\infty$ as $x \to +\infty, f \to -\infty$ as $x \to -\infty$; neither even nor odd.

6. **(88–91) Describe the graph of g based on the basic graph:** $(88) g(x) = x^2 - 1$; $(89) g(x) = (x + 3)^2 + 1$; $(90) g(x) = \sqrt{x} + 2$; $(91) g(x) = -\sqrt{x} - 1$.
   [Source: OpenStax §1.2, exercises 88–91]

   Solution. (88) Parabola $y = x^2$, shifted down by 1. (89) Parabola, shifted left by 3 and up by 1 (vertex $(-3, 1)$). (90) Square root graph, shifted left by 2 (domain $[-2, \infty)$). (91) Square root graph, reflected across the $x-$axis and shifted down by 1 (domain $[0, \infty)$, values ≤ -1). (

7. $(94) f(x) =$ (**4x** $+3, x \leq 0$; $-x + 1, x > 0$): **calculate** $f(-3), f(0), f(2)$; **sketch.**
   [Source: OpenStax §1.2, exercise 94]

   Solution. $f(-3) = 4(-3) + 3 = -9$ (left branch); $f(0) = 3$; $f(2) = -2 + 1 = -1$ (right branch). Graph: increasing line to (0, 3), then decreasing line from open point (0, 1) — jump discontinuity at 0. (

8. $(97) g(x) = (x - 3)/(x - 2),\ x \neq 2$: **calculate** $g(0), g(-4), g(2)$.
   [Source: OpenStax §1.2, exercise 97]

   Solution. $g(0) = (0 - 3)/(0 - 2) = (-3)/(-2) = \frac{3}{2}$; $g(-4) = (-7)/(-6) = \frac{7}{6}$; $g(2) = 4$ by special definition (the formula is undefined there — that is why the piecewise value is given).

9. **True or false (98–103):** $(98) f(x) = \frac{4x+1}{7x-2}$ **is transcendental;** $(99) g(x) =$ **is an odd root function; (100) logarithmic function is algebraic;** $(101) f(x) = x^b$ **with real constant b is an exponential function; (102) domain of even-degree root is all real numbers; (103) company bought equipment for \$20,500, after 3 years worth \$12,300 — find $V(t)$ and interpret.**
   [Source: OpenStax §1.2, exercises 98–103]

   Solution. (98) False: ratio of polynomials is rational, hence algebraic, not transcendental. (99) True: cube roots defined for all reals and function is odd. (100) False: x^b is a power function; logarithms are transcendental. (101) False: exponential has variable in exponent, b^x. (102) False: even-degree roots require non-negative radicand, so domain excludes negatives (odd roots accept all reals).

(103) Linear depreciation: $V(t) = 20500 + \frac{12300-20500}{3}t = 320500 - \frac{8200}{3}t \approx 20500 - 2733.33t$. $V(0) = 20500$ — purchase price; crosses $t-$axis at $t \approx 7.5$ years — zero value moment; $V(5) = 20500 - 41000 \approx \$6833$. 3

6.2 OpenStax §1.5 (Exponential and Logarithmic Functions) — selected exercises solved

1. (239–245) Sketch; specify domain, range, horizontal asymptote: *a.* $f(x) = e^x + 2$**;** *b.* $f(x) = -2^x$**;** *c.* $f(x) = 1 - 2^{-x}$**;** *d.* $f(x) = e^{-x} - 1$**.**

[Source: OpenStax §1.5, exercises 239–245]

Solution. *a.* e^x, shifted up by 2: domain R, range $(2, \infty)$, asymptote $y = 2$. *b.* Reflection across $x-$axis: domain R, range $(-\infty, 0)$, asymptote $y = 0$. *c.* As $x \to +\infty, 2^{-x} \to 0$, so $f \to 1$ from below; as $x \to -\infty, 2^{-x} \to \infty$, so $f \to -\infty$: increasing, range $(-\infty, 1)$, asymptote $y = 1$. *d.* Mirror of e^x (decay right): domain R, range $(-1, \infty)$, asymptote $y = -1$.

d. $\ln 1/e^3 = -3$**.**

[Source: OpenStax §1.5, exercises 246–253]

Solution. By (39): *a.* $3^4 = 81$. *b.* $8^{1/3} = 2$. *c.* $5^0 = 1$. *d.* $e^{-3} = 1/e^3$.

3. (254–262) Rewrite in equivalent logarithmic form: *a.* $2^3 = 8$**;**

b. $4^{-2} = \frac{1}{16}$**;** *c.* $10^2 = 100$**;** *d.* $3^4 = 81$**;** *e.* $3^3 = 27$**;** *f.* $9^y = 150$**.**

[Source: OpenStax §1.5, exercises 254–262]

Solution. *a.* $\log_2 8 = 3$. *b.* $\log_{1/4} 16 = -2$. *c.* $\log 100 = 2$. *d.* $\log_{1/3} x$ means the exponent: $(\frac{1}{3})^y = x$; $\log_{64} 4 = \frac{1}{3}$. *f.* $\log_9 150 = y$.

4. (264–266) Sketch; specify domain, range, vertical asymptote: *a.* $f(x) = 3 + \ln x$**;** *b.* $f(x) = \ln(x - 1)$**;** *c.* $f(x) = \ln(-x)$**.**

[Source: OpenStax §1.5, exercises 264–266]

Solution. *a.* $\ln x$, shifted up by 3: domain $(0, \infty)$, range R, vertical asymptote $x = 0$. *b.* Shift right by 1: domain $(1, \infty)$, range R, asymptote $x = 1$. *c.* Reflection across the $y-$axis: domain $(-\infty, 0)$, range R, asymptote $x = 0$.

d. $\log_5 \sqrt{}$**(125xy**^{3}**).**

[Source: OpenStax §1.5, exercises 270–275]

Solution. *a.* $4 \log x$. *b.* $\log 9 + 3 \log a - \log b = 2 \log 3 + 3 \log a - \log b$. *c.* $\ln a + \frac{1}{3} \ln b$. *d.* $\frac{1}{5} \log_5 (125\text{xy}^3) = 1_5(3 + \log_5 x + 3 \log_5 y)$.

6. (276–283) Solve exactly: *a.* $5^x = 125$**;** *b.* $e^3 x - 15 = 0$**;** *c.* $8^x = 4$**;** *d.* $4^{x+1} - 32 = 0$**;** *e.* $3^{x/14} = \frac{1}{10}$**;** *f.* $10^x = 7.21$**.**

[Source: OpenStax §1.5, exercises 276–283]

Solution. *a.* $x = 3$ (since $125 = 5^3$). *b.* $3\text{x} = \ln 15, x = {}^{ln15}\approx 0.9027$. *c.* $2^3 x = 2^2, 3x = \frac{2}{3}$. *d.* $4^{x+1} = 2^5 \Rightarrow 2^{2x+2} = 2^5 \Rightarrow x = 3\frac{x}{14} \ln 3 = -\ln 10$, $x = -\frac{14ln10}{2}$. *e.* ≈ -29.32. *f.* $\ln 3x = \log 7.21 \approx 0.8580$.

7. (284–291) Solve exactly, if possible: *a.* $\log_3 x = 0$**;** *b.* $\log_5 x = -2$**;** *c.* $\log_3(x+5) = 0$**;** *d.* $\log_2($**2x** $-7) = 0$**;** *e.* $\ln \sqrt{x} + 3 = 2$**;** *f.* $\log_6(x + 9) + \log_6 x = 2$**;** *g.* $\log_4(x + 2) - \log_4(x - 1) = 0$**.**

[Source: OpenStax §1.5, exercises 284–291]

Solution. $a.\ x = 3^0 = 1.\ b.\ x = 5^{-2} = \frac{1}{25}.\ c.\ x + 5 = 3^0 = 1, x = -4$; check domain $x + 5 > 0$: $1 > 0 OK$, so $x = -4$ is valid ($\log_3 1 = 0$). $d.$ 2x $-7 = 1, x = 4.\ e.\ \sqrt{x} + 3 = e^2$, $x = e^4 - 3 \approx 51.60.\ f.$ Combine: $\log_6$

$$\log_6(x(x+9)) = 2$$

$g.\ \log_4(x+2)/(x-1) = 0 \Rightarrow (x+2)/(x-1) = 1 \Rightarrow x+2 = x-1$: impossible; no solutions.

8. (292–297) Change of base, approximation to four decimal places: $a.\ \log_5 47$**;** $b.\ \log_2 82$**;** $c.\ \log_{0.5} 211$**;** $d.\ \log_{0.2} 0.452$**.**

[Source: OpenStax §1.5, exercises 292–297]

Solution. By (41): $a.\ \ln 47/\ln 5 \approx 2.3922.\ b.\ \ln 82/\ln 2 \approx 6.3576.\ c.\ \ln 211/\ln 0.5 \approx -7.7181.\ d.\ \ln 0.452/\ln 0.2 \approx 0.3536.$

9. $(307)A$ **bacterial colony doubles in 12 hours, initially 1000.** $a.$ **Using** $Q = Q_0 e^{kt}$**, find** k**;** $b.$ **how long until 200,000?**

[Source: OpenStax §1.5, exercise 307]

Solution. $a.$ Doubling: $2 = e^{12k}$, so $k = \ln\frac{2}{12} \approx 0.0578$ per hour. $b.\ 200000 = 1000\mathrm{e}^{kt}$: $e^{kt} = 200$, $t = \ln 200/k = 12 \cdot \ln 200/\ln 2 \approx 92.1$ hours.

10. $(308)A$ **rabbit population doubles every 6 months, initially 120.** $a.$ **Using** $P = P_0 a^t$**, find** a**;** $b.$ **when will it reach 3500?**

[Source: OpenStax §1.5, exercise 308]

Solution. $a.\ a = 2$ for $a6-$month period (or per year: with t in years $a = 4$; for 6 months $a = 2$). $b.\ 3500 = 120 \cdot 2^t$ (with t in 6-month units): $2^t = \frac{175}{6} \approx 29.17$, $t = \log_2 29.17 = \ln 29.17/\ln 2 \approx 4.87$ periods, $i.e.$ about 2.4 years.

11. (309) The 1906 San Francisco earthquake had magnitude 8.3; the Japanese earthquake of the same year had magnitude 4.9. How much more intense was San Francisco (Richter scale)?

[Source: OpenStax §1.5, exercise 309]

Solution. Ratio of Richter intensities: $10^{8.3-4.9} = 10^{3.4} \approx 2512$: the San Francisco earthquake was about 2500 times more intense — a vivid demonstration of the logarithmic scale from (40).

6.3 OpenStax §2.2 — table exercises (42–45) solved

1. Estimate $\lim_{t\to 0+} \cos t/t$ **by table** $(t = 0.1,\ 0.01,\ 0.001,\ 0.0001)$**.**

[Source: OpenStax §2.2, exercise 42]

Solution. Values: $t = 0.1$: $0.\frac{995004}{0}.1 = 9.95004$; $t = 0.01$: $0.\frac{99995}{0}.01 = 99.995$; $t = 0.001$: $0.9999995/0.001 = 999.9995$; $t = 0.0001$: 9999.95. Values grow without bound: the numerator $\to 1$ while the denominator $\to 0^+$ — $aK/0^+$ form, so the limit is $+\infty$.

$$\lim_{t\to 0+} \cos(t)/t = +\infty$$

2. Estimate $\lim_{x\to 2}(x-2)/(x^3 -$ **4x) by table.**

[Source: OpenStax §2.2, exercise 43]

Solution. First simplify: $(x-2)/(x^3 - 4\mathrm{x}) = (x-2)/(x(x-2)(x+2)) = 1/(x(x+2))$ for $x \neq 2$. Table values: $x = 1.99$: $1/(1.99 \cdot 3.99) = \frac{1}{7.9401} \approx 0.1259$; $x = 2.01$: $1/(2.01 \cdot 4.01) =$

$\frac{1}{8}.0601 \approx 0.1241$ — approaching $0.125 = \frac{1}{8}$. Algebra confirms the table: the limit is $1/(2 \cdot 4) = \frac{1}{8}$. T Guess from the table, then verify by the graph: $\lim_{\theta \to 0} \sin(\pi/\theta)$.

[Source: OpenStax §2.2, ex. 44]

Solution. Values: $\theta = 0.1$: $\sin(10\pi) = 0$; 0.01: $\sin(100\pi) = 0$; 0.001: $\sin(1000\pi) = 0$; 0.0001: 0. The table suggests 0 — but $\sin(\pi/\theta)$ oscillates between -1 and 1 infinitely fast near 0 (for example, $\theta = \frac{2}{3}$ gives $\sin(3\pi/2) = -1$, and $\theta = \frac{2}{5}$ gives $\sin(5\pi/2) = 1$). The limit does not exist; the table was misleading because the sample points $1/n$ were exactly zeros of sine. This is a cautionary exercise about the tabular method. T Guess from the table, then verify by the graph: $\lim_{\alpha \to 0+}(1/\alpha) \cdot \cos(\pi/\alpha)$.

[Source: OpenStax §2.2, ex. 45]

Solution. When $\alpha = 1/n$: $(1/\alpha) \cdot \cos(n\pi) = n \cdot (-1)^n$ — the values $\pm n$ grow in magnitude and alternate in sign. The graph oscillates between ever larger positive and negative values: the limit does not exist (neither $+\infty$ nor $-\infty$ — the signs alternate). The table at lucky points ($\alpha = 1/(2\text{n})$ gives $+2\text{n} \to +\infty$) would have been misleading in the opposite direction.